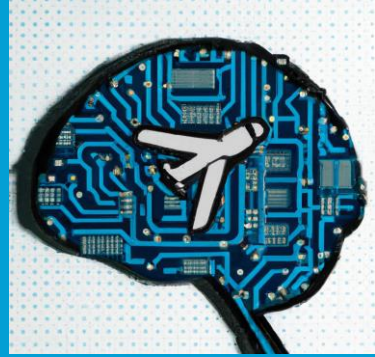
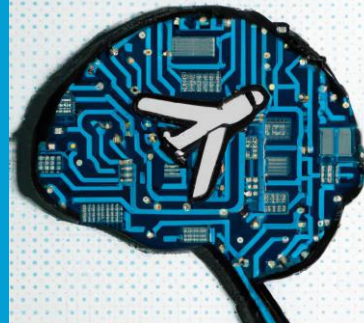




Understanding (deep) neural networks

Lecture objectives

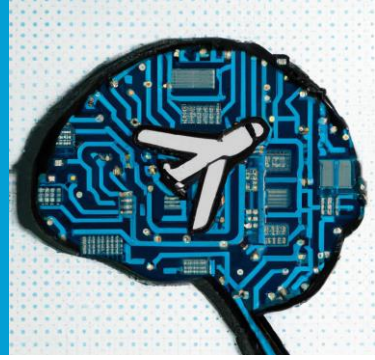


Students can:

- Motivate the need for understanding (deep) neural networks and for explainable AI
- Explain what it means to “understand” AI
- List, categorize, explain, and apply methods that are used for analyzing (deep) neural networks



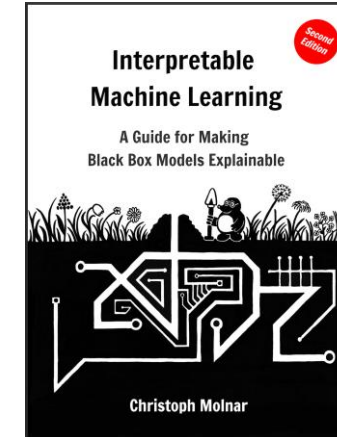
Recommended literature



Interpretable Machine Learning

A Guide for Making Black Box Models Explainable

By Christoph Molnar



<https://christophm.github.io/interpretable-ml-book/index.html>

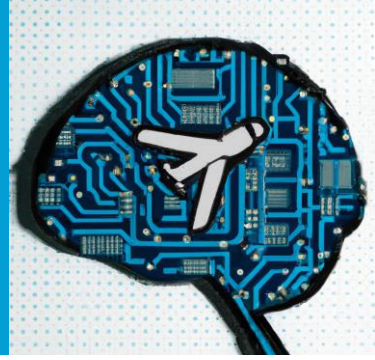
Interactive reader on github:

https://github.com/guidoAI/understanding_neural_networks

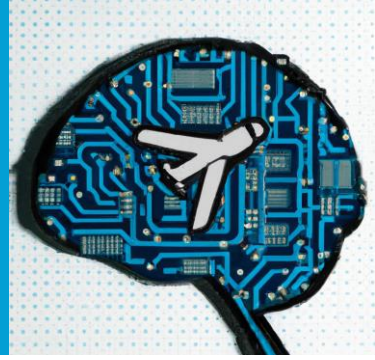
https://mybinder.org/v2/gh/guidoAI/understanding_neural_networks/HEAD?urlpath=%2Fdoc%2Ftree%2FUnderstanding+neural+networks.ipynb



Why understand deep neural networks?

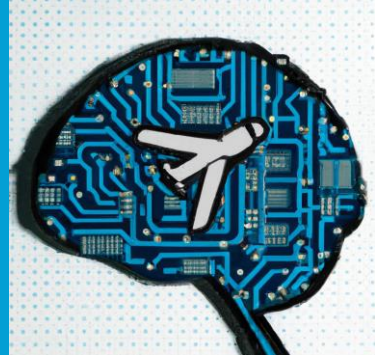


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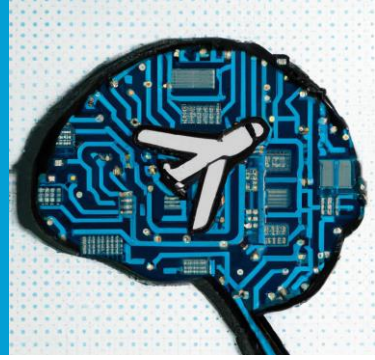
1. Justifying outcomes

Why understand deep neural networks?



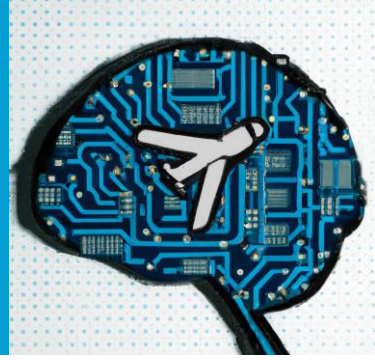
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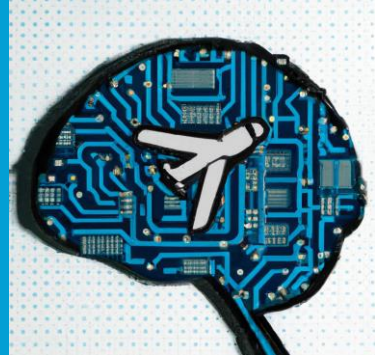
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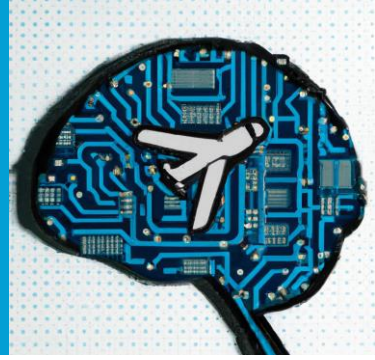
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4. Learning from the network's solution

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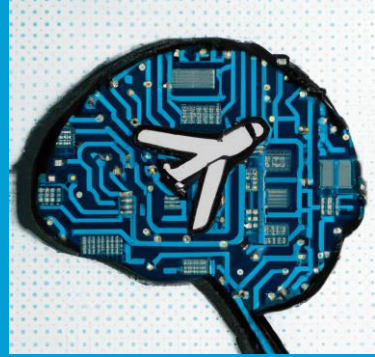


1. Justifying outcomes
2. Predicting the network's outputs for unseen samples
3. Improving the network (e.g., its training)
4. Learning from the network's solution
5. The inherent satisfaction of understanding

EU guidelines on ethics in AI

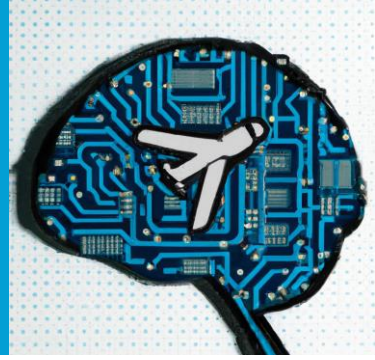


- Human agency and oversight
- Robustness and safety
- Privacy and data governance
- Transparency
- Diversity, nondiscrimination and fairness
- Societal and environmental well-being
- Accountability

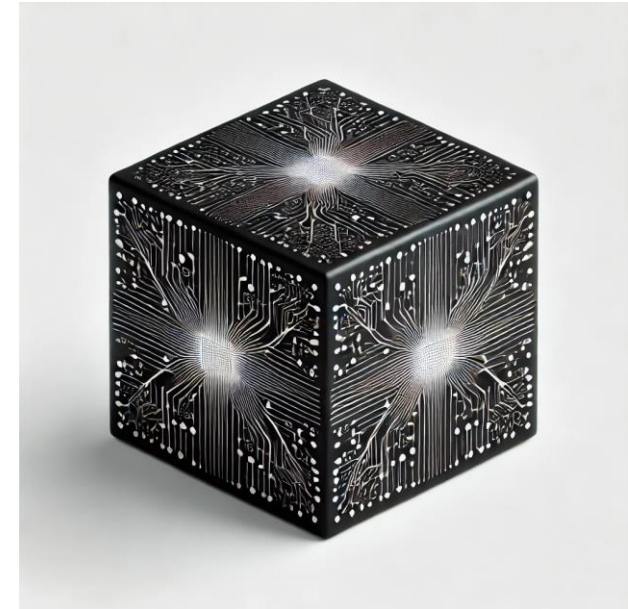


The problem

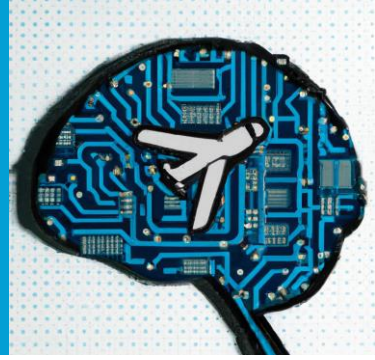
What is the problem?



Deep neural networks are often called “black boxes”

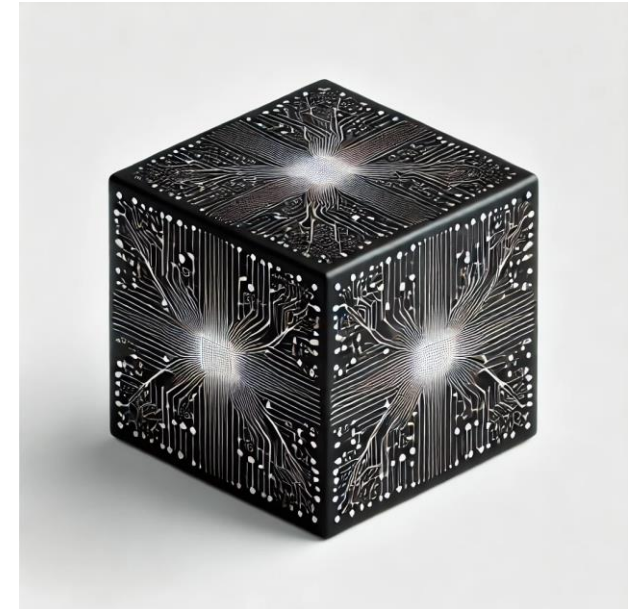


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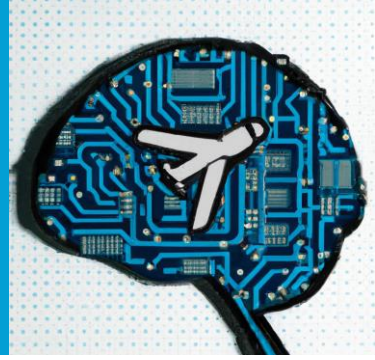


Deep neural networks are often called “black boxes”

Would you agree with this?

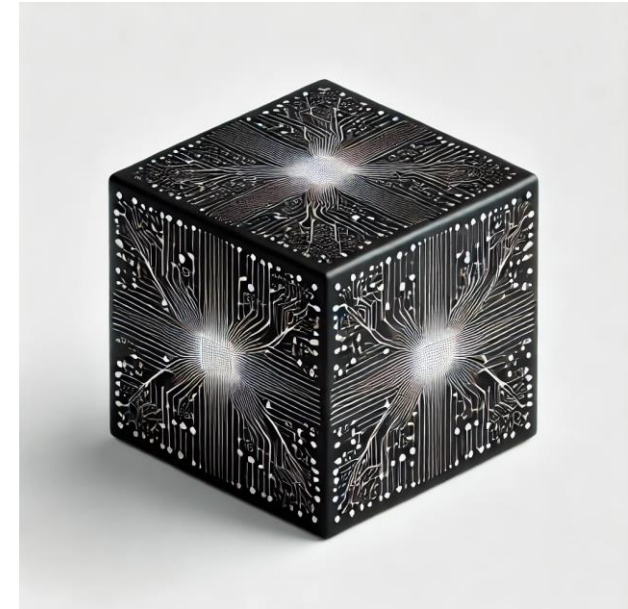


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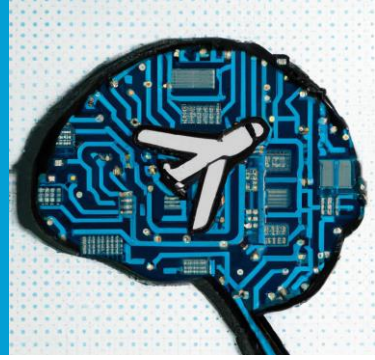


Deep neural networks are often called “black boxes”

It suggests that they are not to be understood 🙄



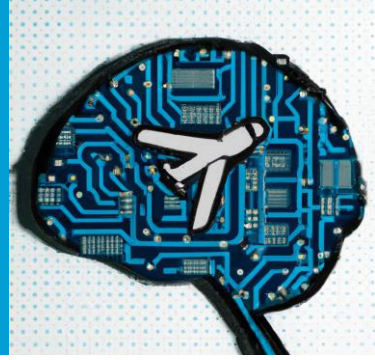
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Deep neural networks are actually *white* boxes



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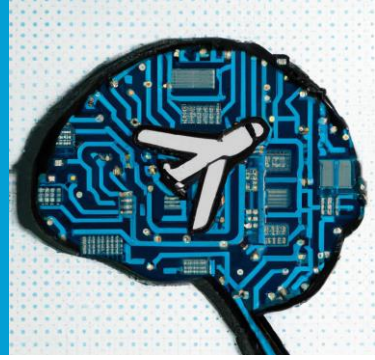


Deep neural networks are actually *white* boxes

We know ***exactly*** what is going on inside the box!



What is the problem?



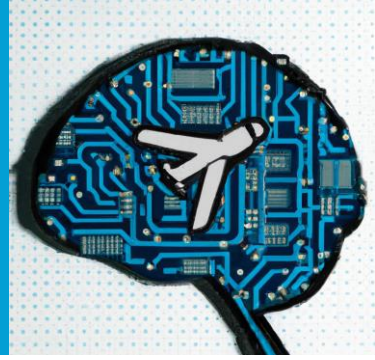
Deep neural networks are actually *white* boxes

We know ***exactly*** what is going on inside the box:

- The computer can only execute a network, since all operations are known and well defined.



What is the problem?



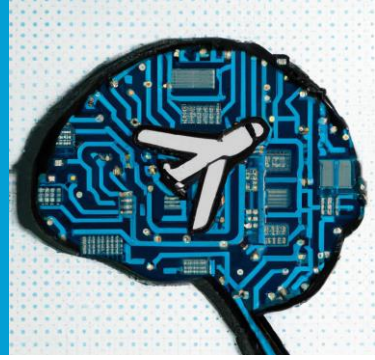
Deep neural networks are actually *white* boxes

We know ***exactly*** what is going on inside the box:

- The computer can only execute a network, since all operations are known and well defined.
- We can track and manipulate any part of the network in any way we want. Compare that to real animal brains 🧠



What is the problem?



Deep neural networks are actually *white* boxes

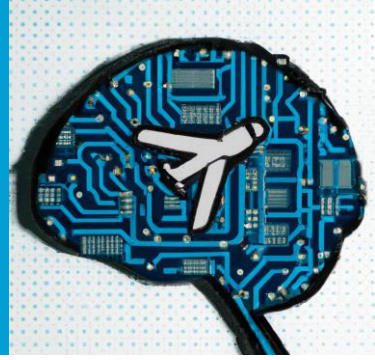
We know ***exactly*** what is going on inside the box:

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So, what is the problem then?



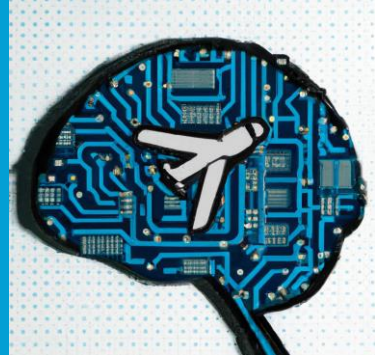
What is the problem?



The problem comes from **complexity**:



What is the problem?

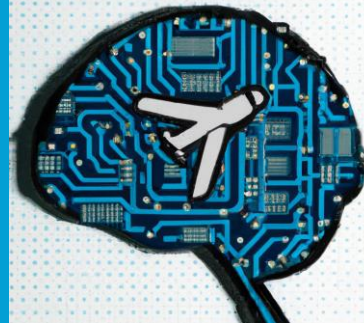


The problem comes from **complexity**:

“Complexity characterizes the behavior of a system or model whose components interact in multiple ways and follow local rules, leading to non-linearity, randomness, collective dynamics, hierarchy, and emergence.”



What is the problem?



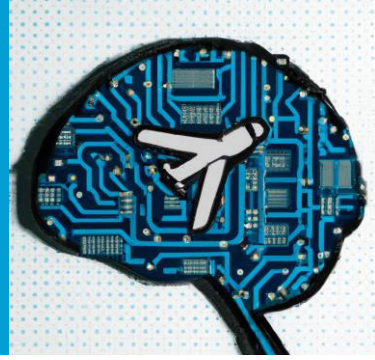
The problem comes from **complexity**:

“Complexity characterizes the behavior of a system or model whose components interact in multiple ways and follow local rules, leading to non-linearity, randomness, collective dynamics, hierarchy, and emergence.”

“Complex systems are systems whose behavior is intrinsically difficult to model due to the dependencies, competitions, relationships, or other types of interactions between their parts or between a given system and its environment.”



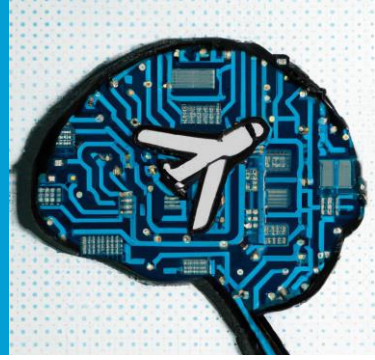
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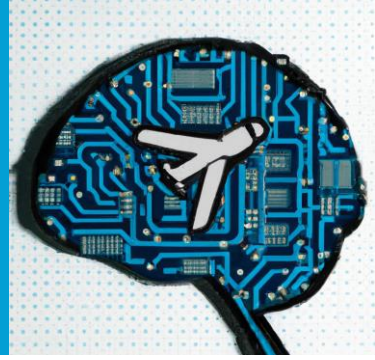
In the context of deep neural networks:

Although we know and understand all **local** components, we do not understand the **global** behavior of the network.

What does it mean to “understand”?



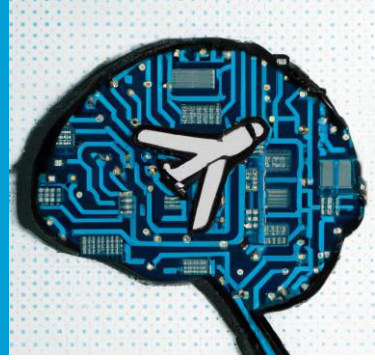
What does it mean to “understand”?



Definitions of “understanding” and “knowledge” are under heavy debate.



What does it mean to “understand”?

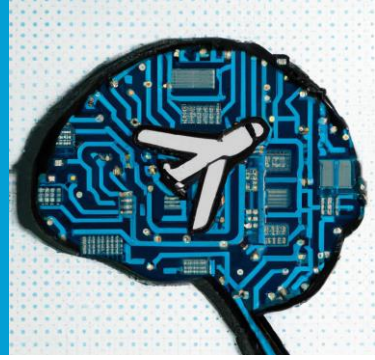


Definitions of “understanding” and “knowledge” are under heavy debate.

What do you think it means to “understand” something?



What does it mean to “understand”?

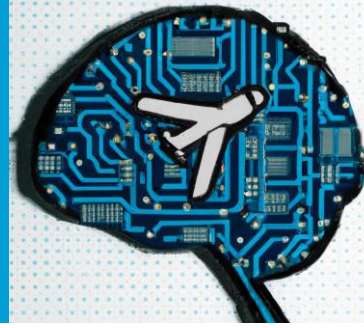


Definitions of “understanding” and “knowledge” are under heavy debate.

Knowledge is typically associated with knowing a single fact.



What does it mean to “understand”?



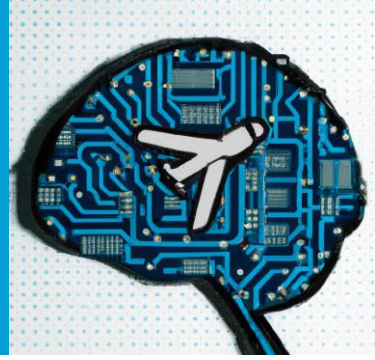
Definitions of “understanding” and “knowledge” are under heavy debate.

Knowledge is typically associated with knowing a single fact.

Understanding is generally linked to the relations between facts in a body of knowledge, allowing for prediction and control.



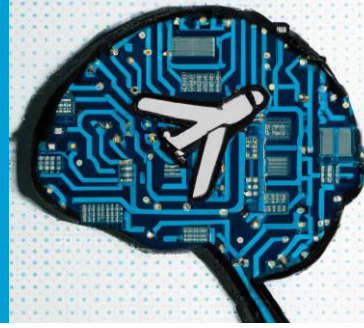
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(Objective) understanding is tightly linked to the scientific process of forming models / hypotheses and testing them by means of verifying predictions.



What does it mean to “understand”?

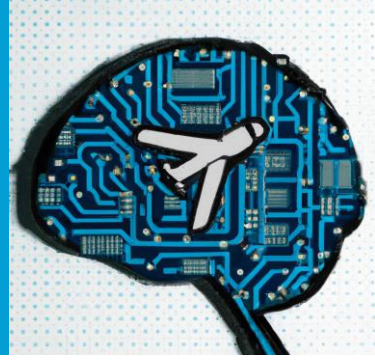


(Objective) understanding is tightly linked to the scientific process of forming models / hypotheses and testing them by means of verifying predictions.

Understanding hence contains an element of compression, e.g., from many data points to a simplified model consisting of a few rules.



What do we want to understand?



Local interpretability

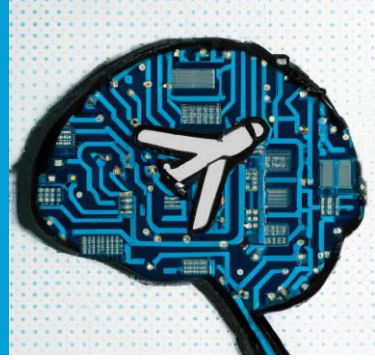
How does a deep neural network arrive at the output for a given specific input?

Global interpretability

How does the neural network generate its outputs for any input?



What do we want to understand?



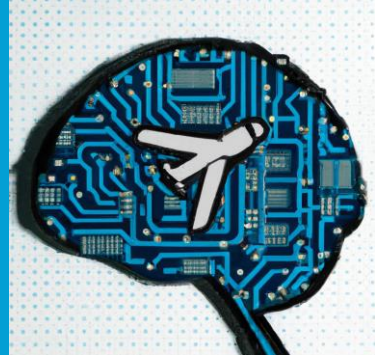
Levels of abstraction for global interpretability

- What strategy does the network employ to solve a task?



“Black box” approach

What do we want to understand?



Levels of abstraction for global interpretability

- What strategy does the network employ to solve a task?
- How does the neural network implement the strategy?

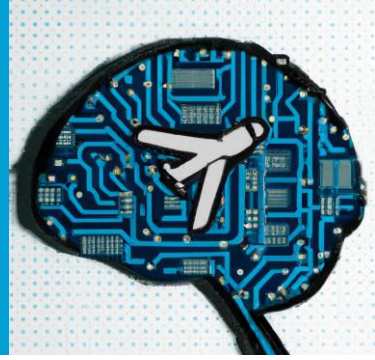


“Black box” approach



“White box” approach

What do we want to understand?



Levels of abstraction for global interpretability

- What strategy does the network employ to solve a task?
- How does the neural network implement the strategy?



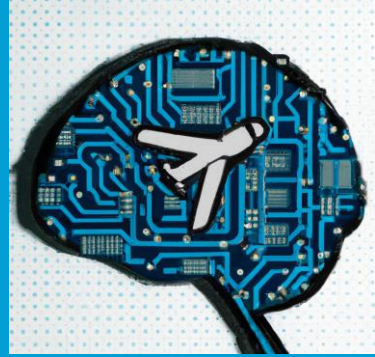
“Black box” approach



“White box” approach

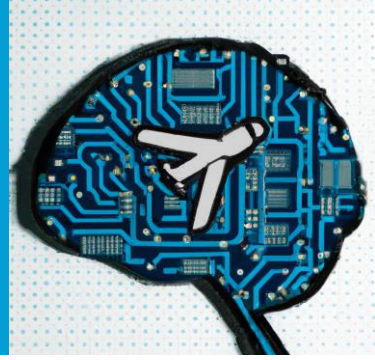
Note:

The “what” and “how” are of course tightly linked.



Methods for interpreting AI

Types of interpretability methods



Intrinsic

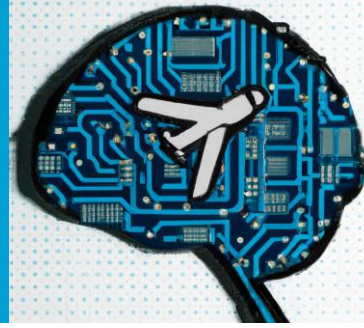
The model is set up so that the parameters are easy to interpret.

Post hoc

Analyze a complex, trained model.

VS.

Types of interpretability methods



Intrinsic

The model is set up so that the parameters are easy to interpret.

Model-agnostic

Can be applied to any model.

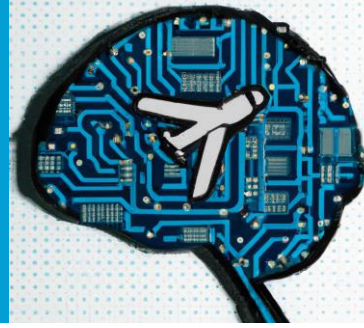
VS.

Post hoc

Analyze a complex, trained model.

Model-specific

Types of interpretability methods



Intrinsic

The model is set up so that the parameters are easy to interpret.

Model-agnostic

Can be applied to any model.

Local

Focus on a single input.

VS.

Post hoc

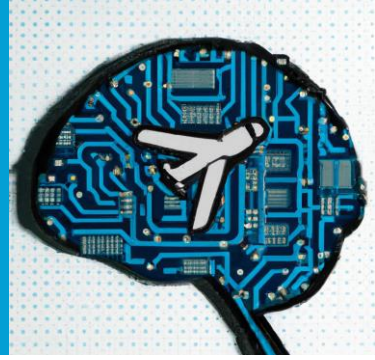
Analyze a complex, trained model.

Model-specific

Global

Predict the behavior for all inputs

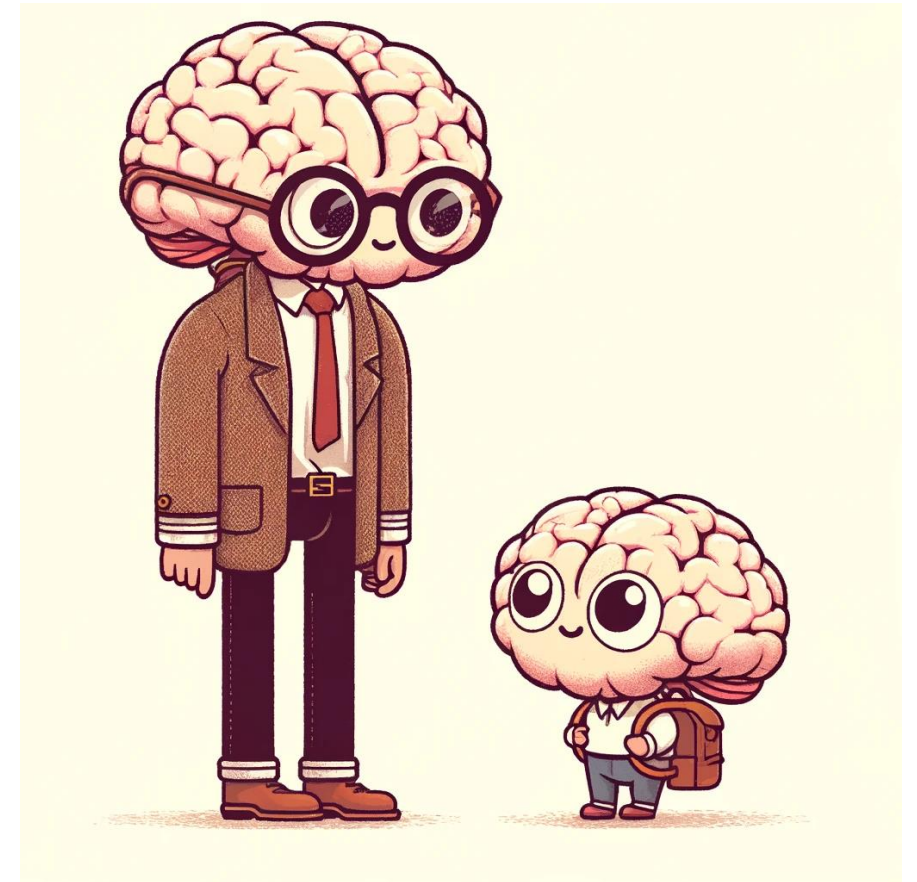
Surrogate models



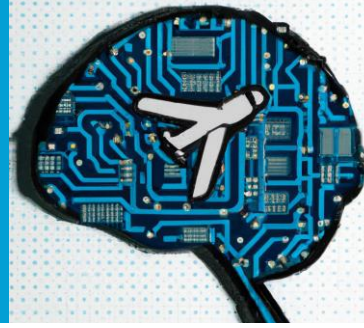
Train a simpler / intrinsically interpretable model to mimic the deep neural network.

For example:

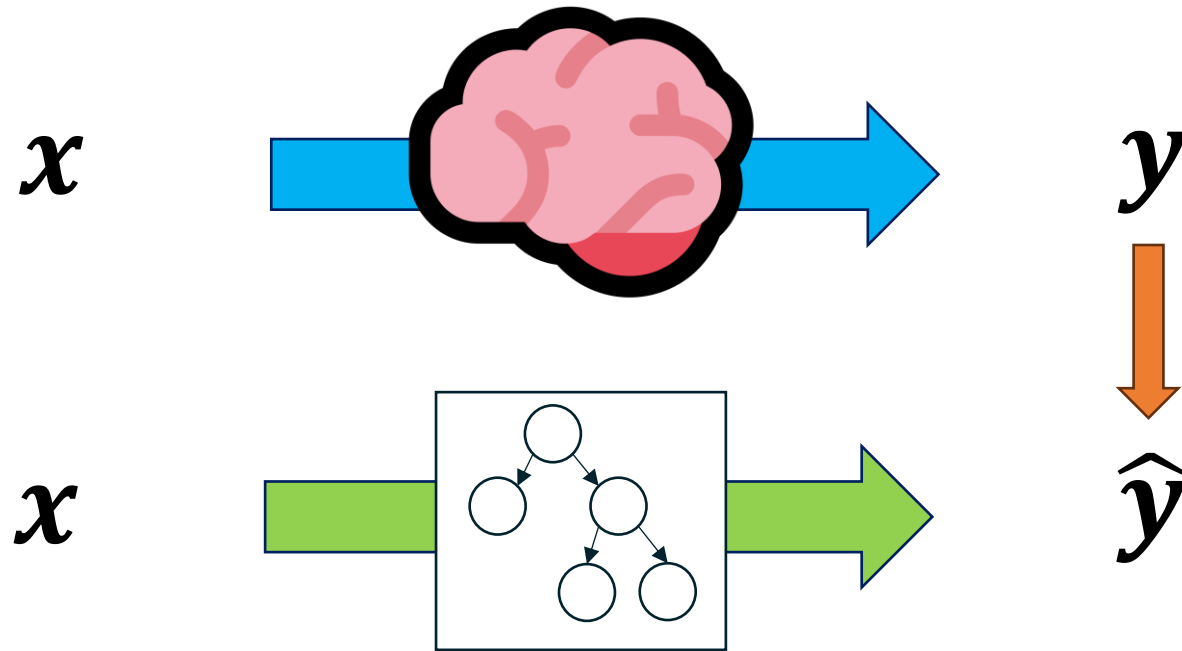
Train a decision tree to mimic the network's input-output mapping.



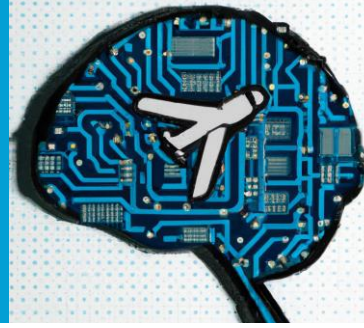
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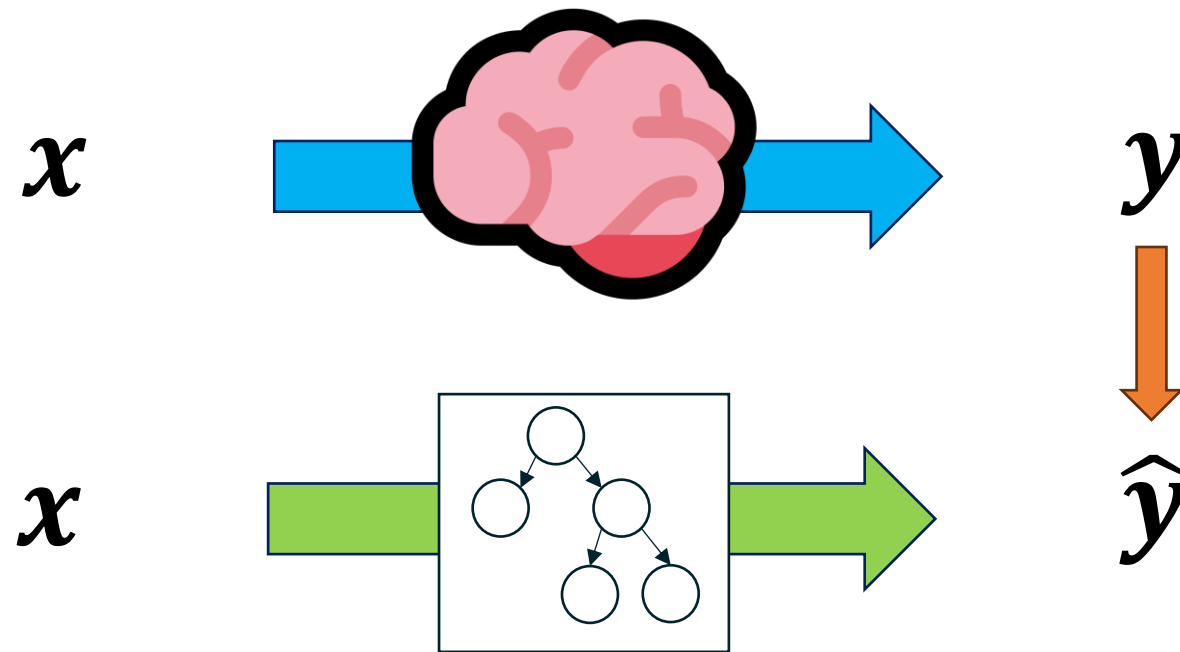
Training procedure:



Surrogate models



Training procedure:

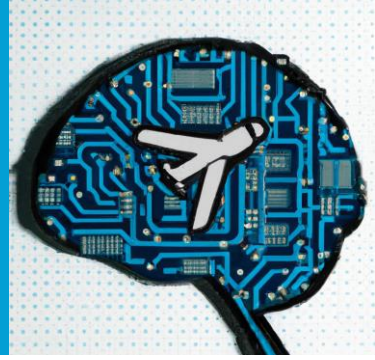


Note:

Different choices can be made for x : it can be the entire training set, or only samples close to a specific sample.

This will determine if the surrogate model is a global / local model.

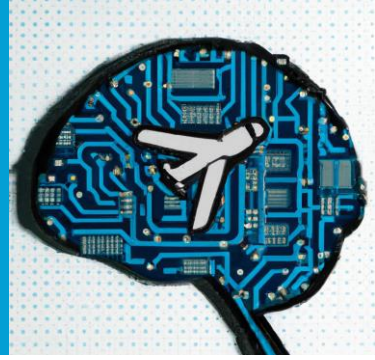
Surrogate model



Advantage(s):

Disadvantage(s):

Surrogate model

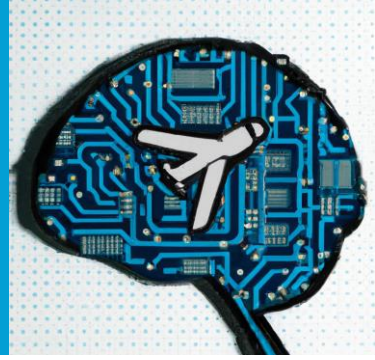


Advantage(s):

- ✓ Flexible, model-agnostic method.
- ✓ If performing well, it may explain certain aspects of the model's behavior.

Disadvantage(s):

Surrogate model



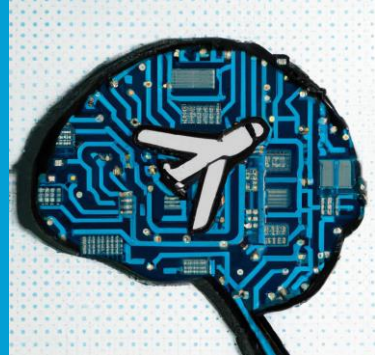
Advantage(s):

- ✓ Flexible, model-agnostic method.
- ✓ If performing well, it may explain certain aspects of the model's behavior.

Disadvantage(s):

- ✗ Globally, the model will typically perform (much?) less well than the complex deep neural network. *Trade-off accuracy and interpretability.*
- ✗ The behavior will typically not be exactly the same, so it leaves aspects unexplained.

LIME



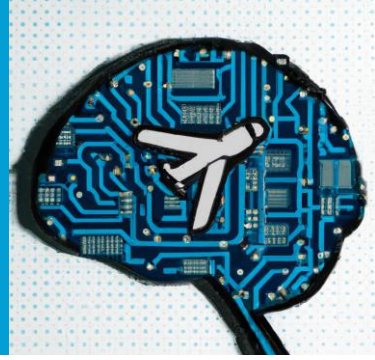
Local interpretable model-agnostic explanations (LIME)

Key ideas:

- Explain a single sample decision by training a local surrogate model



LIME



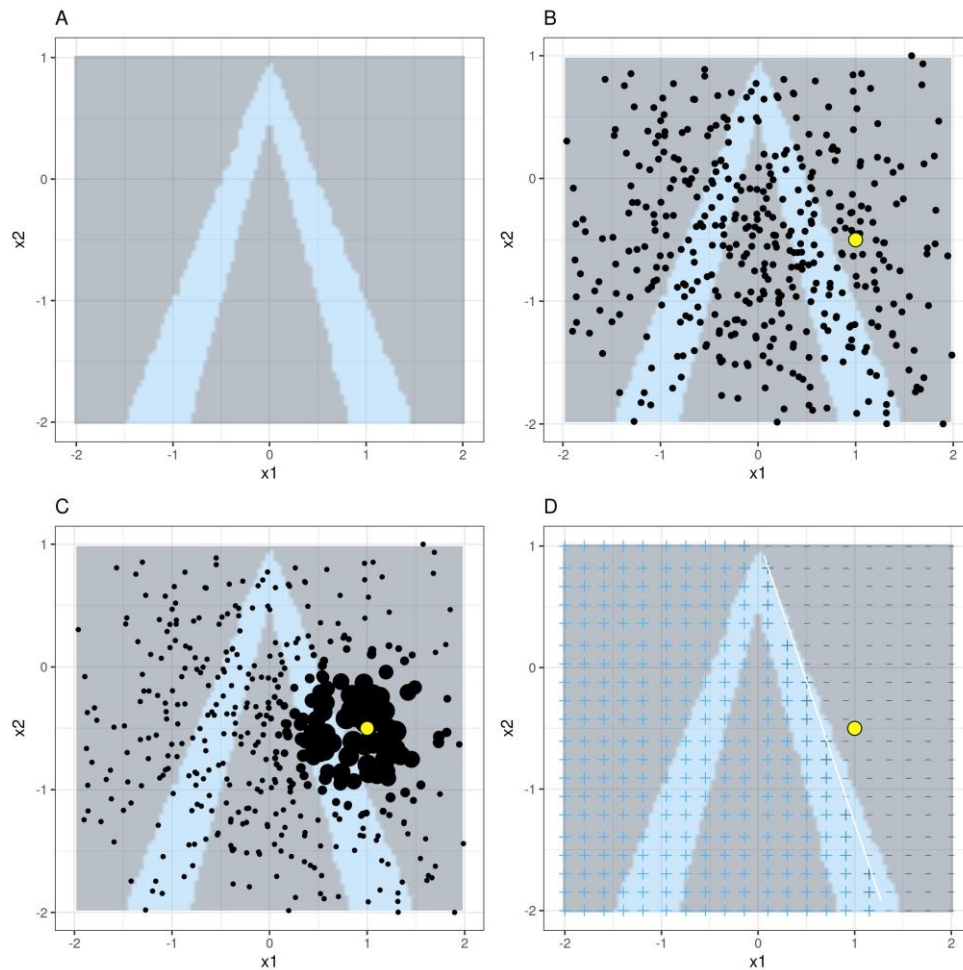
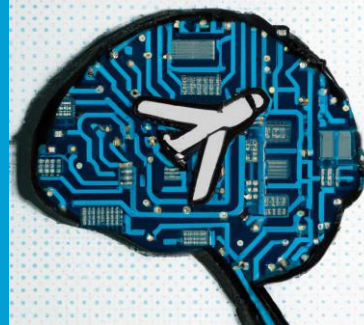
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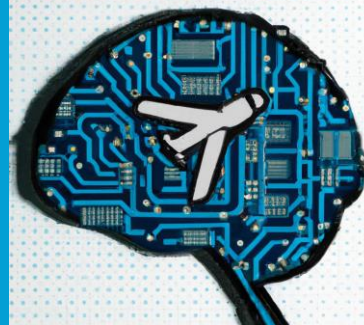
- Explain a single sample decision by training a local surrogate model
- Generate training samples by perturbing the given sample, assigning larger weights to closer samples



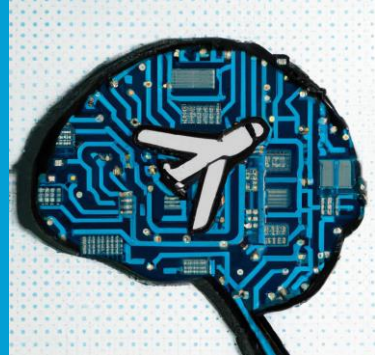
LIME



LIME



LIME



“Inherits” the advantages & disadvantages of surrogate models, plus:

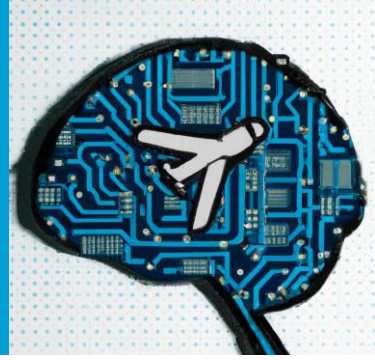
Advantage(s):



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LIME



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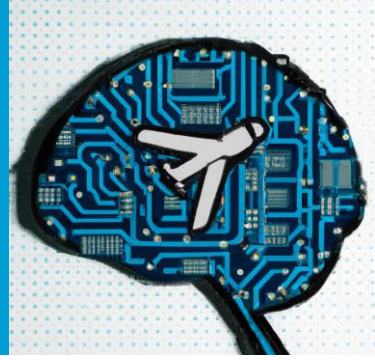
Advantage(s):

✓ Can allow for “contrastive” explanations, as perturbations may cause the class to change.

Disadvantage(s):

- ✗ The right “neighborhood” of perturbation is hard to define.
- ✗ Perturbations may lead to infeasible samples.

Shapley values



Concept from coalitional game theory:

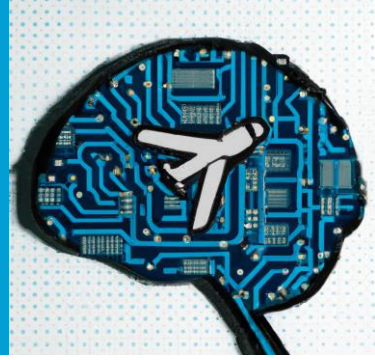
- How much does a specific feature contribute to the outcome of a specific input sample?

Input sample: $\mathbf{x} = \langle x_1, x_2, \dots, x_n \rangle$

What is the influence of feature i , that is x_i ?



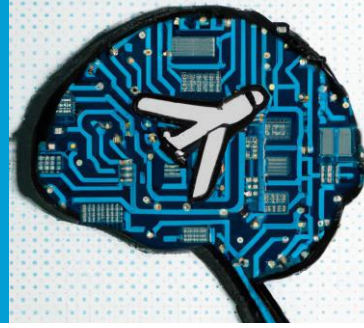
Shapley values



Key ideas:

- Generate many different input samples that contain some of the original input feature values (x_i). The values of the other features are sampled from the data set (X_i).
- Determine the contribution of x_i to the result $f(x_i)$.

Shapley values



Approximation with Monte Carlo sampling:

$$\varphi_i = \frac{1}{M} \sum_{m=1}^M \left(f(x_{+i}^m) - f(x_{-i}^m) \right)$$

Input sample:

$$x = \langle x_1, x_2, x_3, x_4 \rangle$$

Generated samples:

$$x_{+i}^1 = \langle x_1, z_2, x_3, x_4 \rangle$$

$$x_{-i}^1 = \langle z_1, z_2, x_3, x_4 \rangle$$

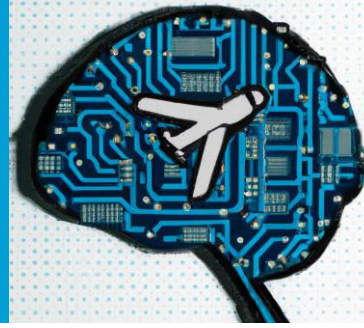
$$x_{+i}^2 = \langle x_1, x_2, z_3, z_4 \rangle$$

$$x_{-i}^2 = \langle z_1, x_2, z_3, z_4 \rangle$$

...

With $z_j \sim X_j$.

Shapley values



Approximation with Monte Carlo sampling:

$$\varphi_i = \frac{1}{M} \sum_{m=1}^M \left(f(x_{+i}^m) - f(x_{-i}^m) \right)$$

Note:

In this example, we show the samples for $i = 1$.

Input sample:

$$x = \langle x_1, x_2, x_3, x_4 \rangle$$

Generated samples:

$$x_{+i}^1 = \langle x_1, z_2, x_3, x_4 \rangle$$

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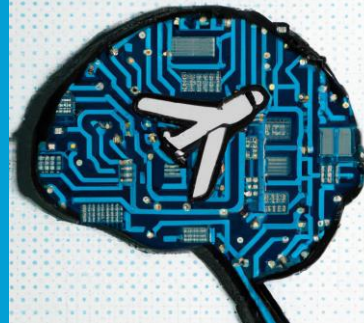
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Shapley values



Approximation with Monte Carlo sampling:

$$\varphi_i = \frac{1}{M} \sum_{m=1}^M \left(f(x_{+i}^m) - f(x_{-i}^m) \right)$$

Note:

In this example, we show the samples for $i = 1$.

Input sample:

$$x = \langle x_1, x_2, x_3, x_4 \rangle$$

Generated samples:

$$x_{+i}^1 = \langle x_1, z_2, x_3, x_4 \rangle$$

$$x_{-i}^1 = \langle z_1, z_2, x_3, x_4 \rangle$$

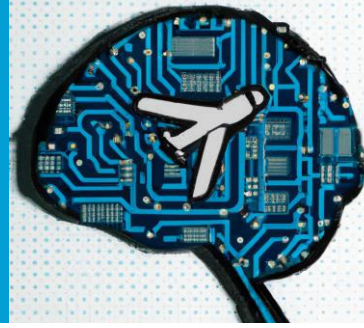
$$x_{+i}^2 = \langle x_1, x_2, z_3, z_4 \rangle$$

$$x_{-i}^2 = \langle z_1, x_2, z_3, z_4 \rangle$$

...

With $z_j \sim X_j$.

Shapley values



Theoretical properties

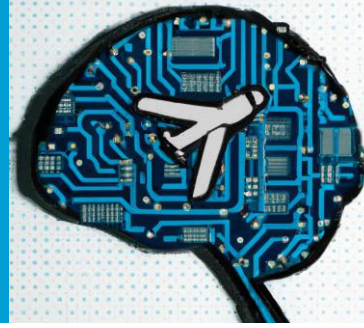
Efficiency:
$$\sum_{i=1}^n \varphi_i = f(\mathbf{x}) - E_X[f(X)]$$

Dummy:
$$\forall S: v(S) = v(S \cup i) \Leftrightarrow \varphi_i = 0$$

Additivity:
$$f(\mathbf{x}) = g(\mathbf{x}) + h(\mathbf{x}) \Rightarrow \varphi_i = \varphi_i^g + \varphi_i^h$$

Symmetry:
$$\forall S \subseteq \{1, 2, \dots, n\} \setminus \{i, j\}: v(S \cup i) = v(S \cup j) \\ \Rightarrow \varphi_i = \varphi_j$$

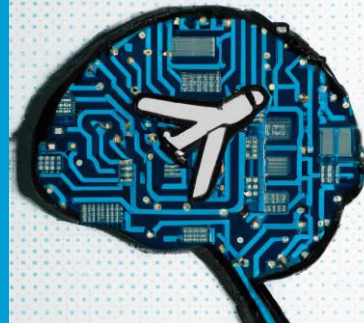
Shapley values



Advantage(s):

Disadvantage(s):

Shapley values



Advantage(s):

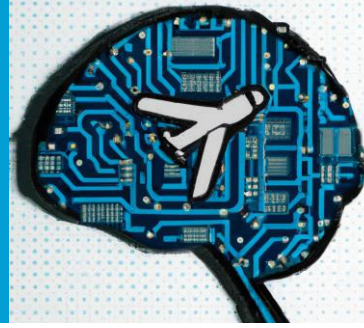
✓ Desirable theoretical properties, notably fair distribution of the difference between $f(x)$ and $E_X[f(x)]$ over the features.

Disadvantage(s):

✗ It takes a lot of computation time – typically only the approximation is possible.

✗ The sampling can lead to the inclusion of unrealistic samples due to ignoring correlations between the features.

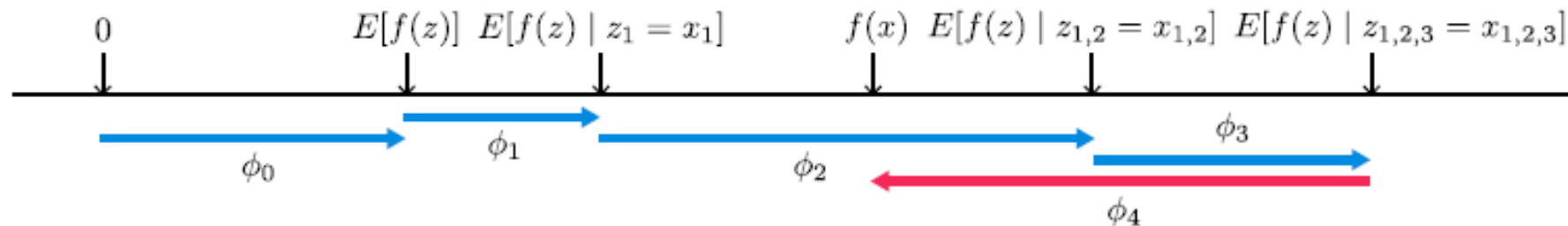
SHAP (SHapley Additive exPlanations)



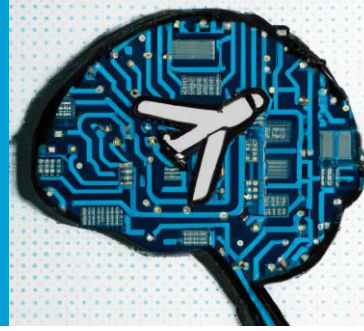
A combination of Shapley values and LIME that estimates a linear model:

$$g(\mathbf{z}) = \varphi_0 + \sum_{i=1}^n \varphi_i z_i$$

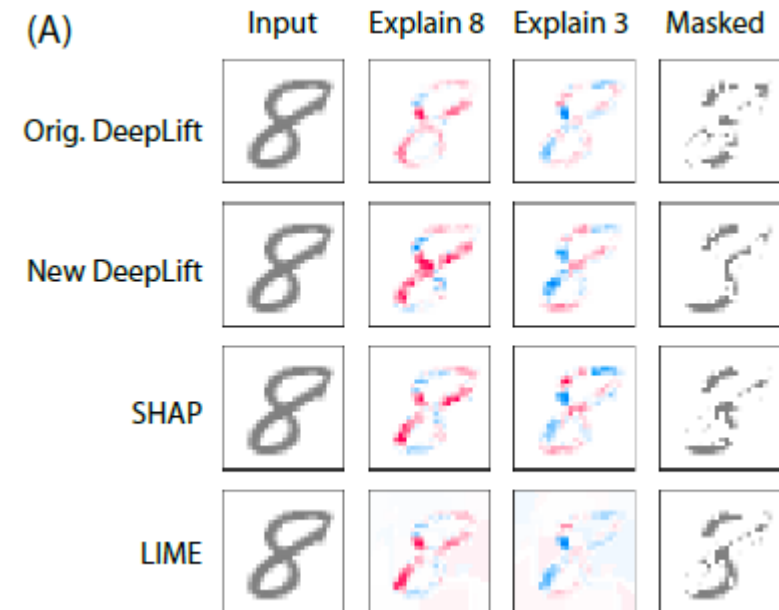
Where $\mathbf{z} = \{0,1\}^n$ is the “coalition vector”, determining which features come from the original sample and which are sampled.



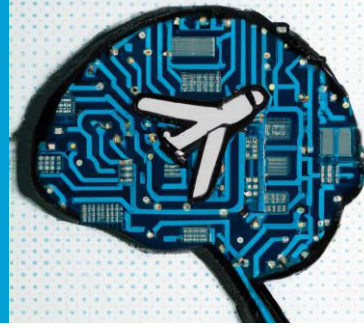
SHAP



Example application to images

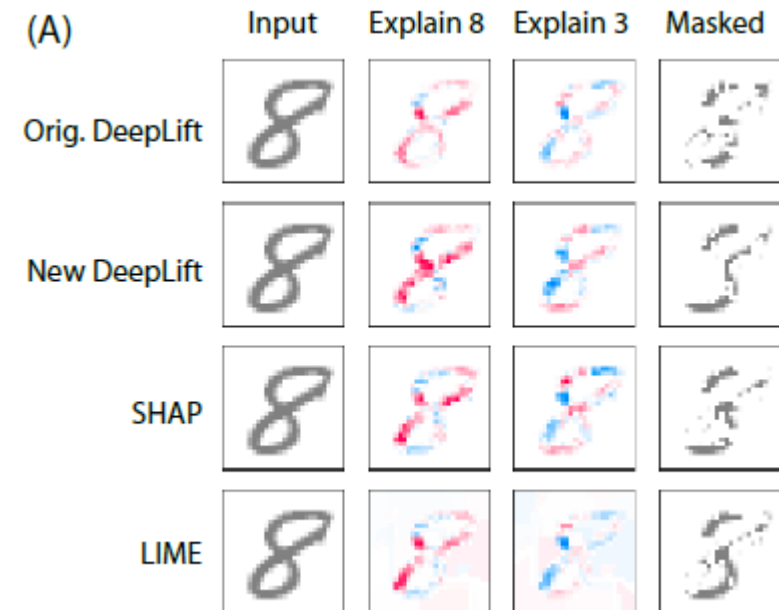


SHAP

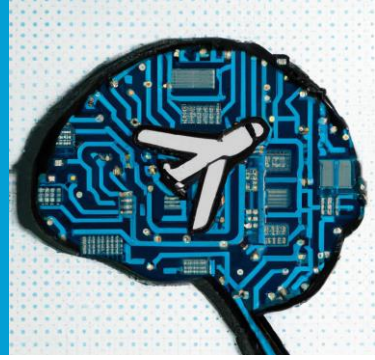


Example application to images

What does this explain to you?



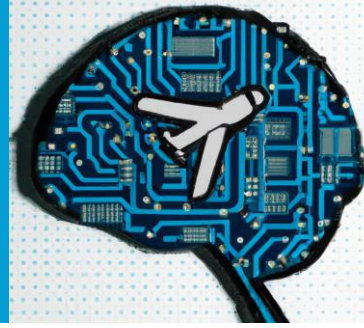
SHAP



Advantages

Disadvantages

SHAP



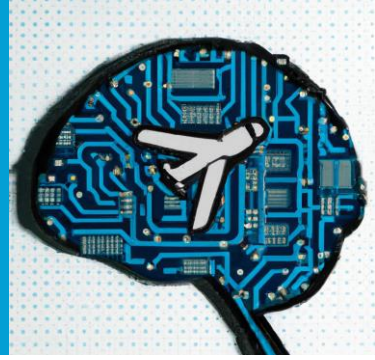
Advantages

- ✓ Solid theoretical foundation
- ✓ Allows for contrastive explanations

Disadvantages

- ✗ Computationally expensive (slow)
- ✗ As other permutation-based methods, ignores feature dependence

Vanilla gradient



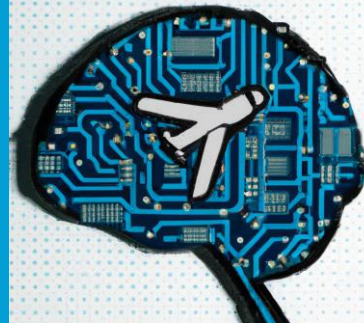
Procedure:

- Perform a forward pass, e.g., for an image I
- Calculate the gradient (as in backprop) from an output of interest o with respect to the input image I (instead of the weights)
- Visualize the gradient



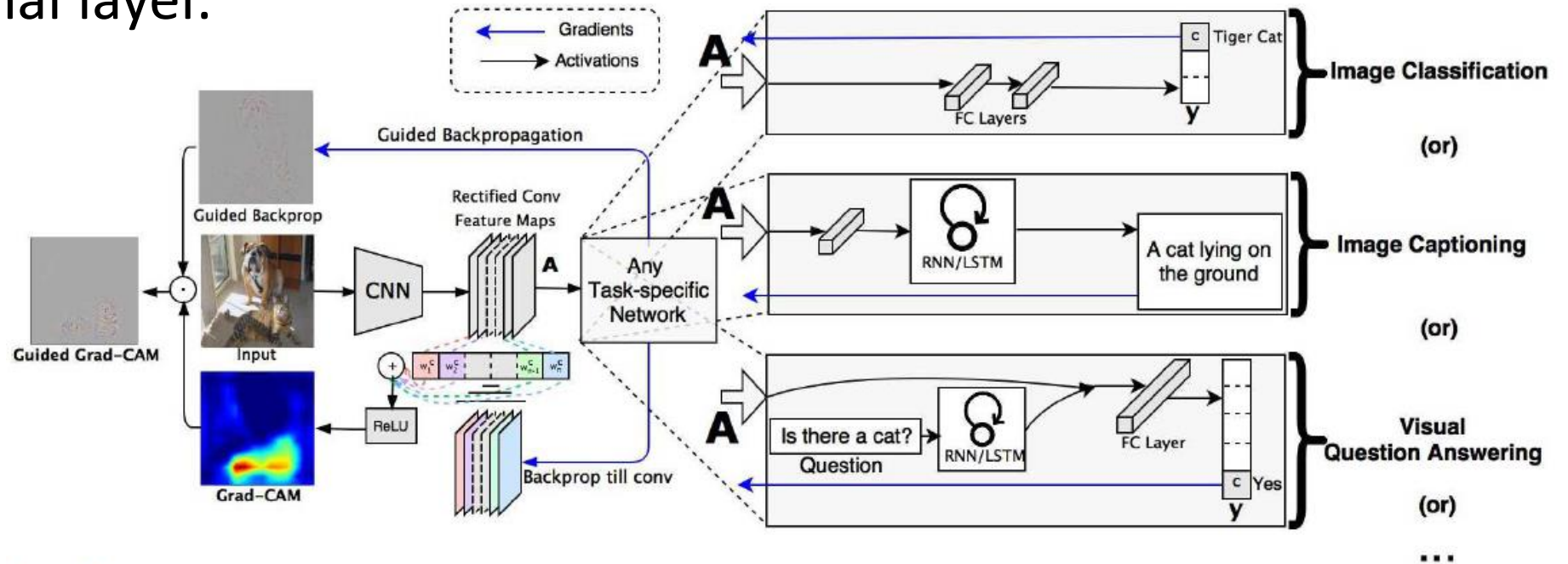
Saliency map: magnitude of the gradient.

Grad-CAM

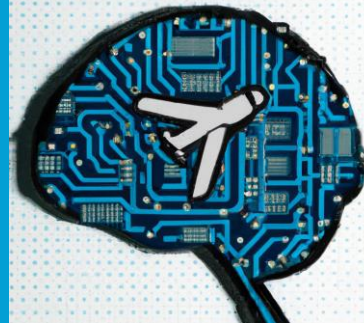


Gradient-weighted Class Activation Mapping

- Focuses on the last activations of and gradients to the last convolutional layer.

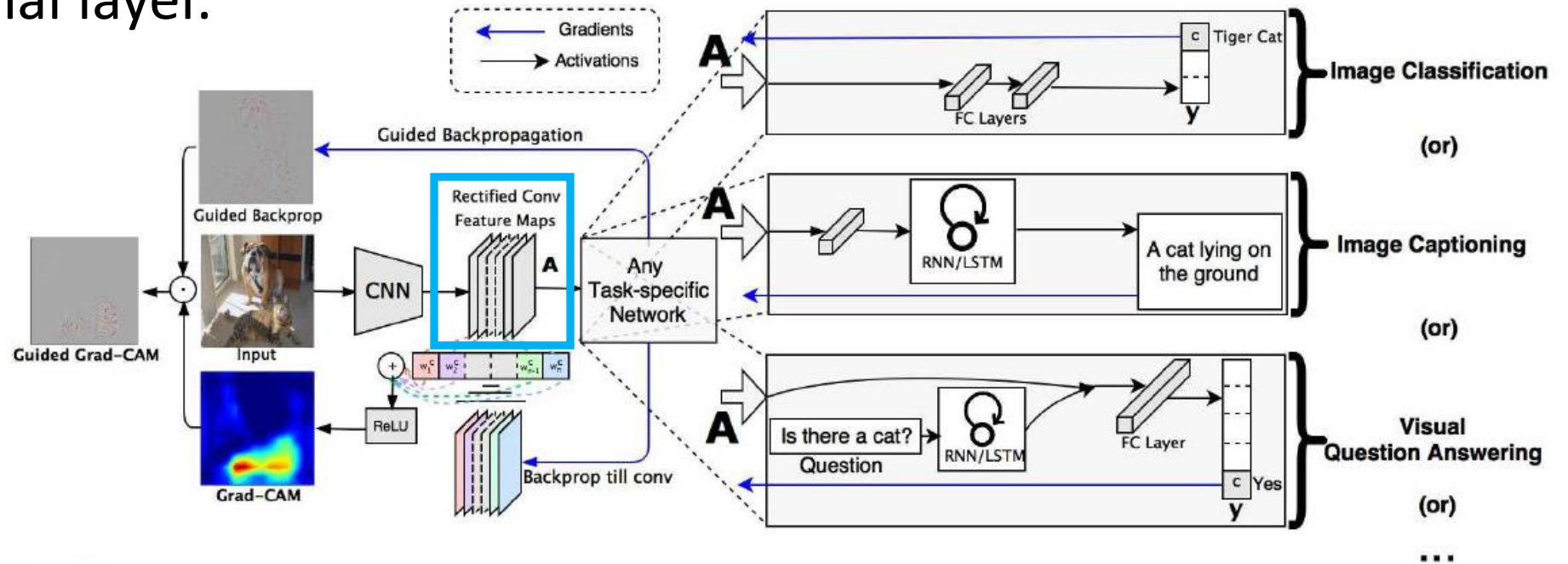


Grad-CAM

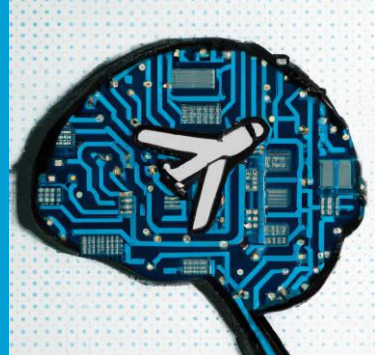


Gradient-weighted Class Activation Mapping

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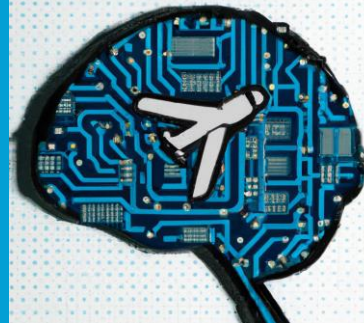


Grad-CAM



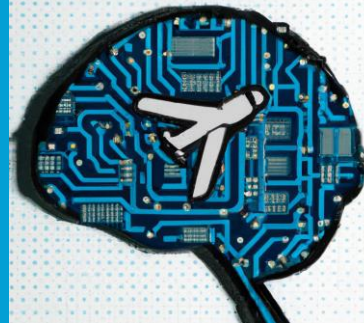
- Calculate the gradient from the raw classification output y^c to the feature channels A^k (of dimension $u \times v$): $\frac{\delta y^c}{\delta A^k}$

Grad-CAM



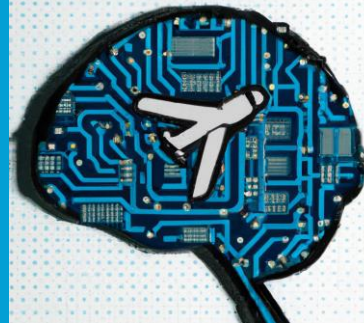
- Calculate the gradient from the raw classification output y^c to the feature channels A^k (of dimension $u \times v$): $\frac{\delta y^c}{\delta A^k}$
- Pool the gradients per channel: $\alpha_c^k = \frac{1}{Z} \sum_i \sum_j \frac{\delta y^c}{\delta A_{i,j}^k}$

Grad-CAM



- Calculate the gradient from the raw classification output y^c to the feature channels $\mathbf{A}^k$ (of dimension $u \times v$): $\frac{\delta y^c}{\delta \mathbf{A}^k}$
- Pool the gradients per channel: $\alpha_c^k = \frac{1}{Z} \sum_i \sum_j \frac{\delta y^c}{\delta A_{i,j}^k}$
- Weigh the activations with α_c^k : $L_c = \text{ReLU}(\sum_k \alpha_c^k \mathbf{A}^k)$

Grad-CAM

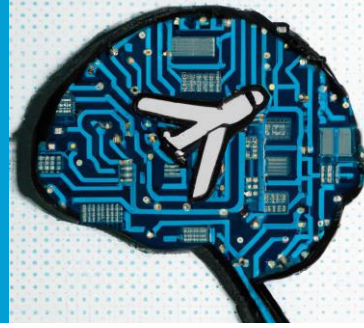


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Guided Grad-CAM:

- Pointwise multiplication of guided backpropagation (e.g., vanilla gradient) with upscaled Grad-CAM.

Grad-CAM



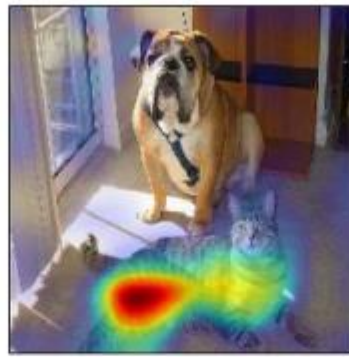
Results on image classification:



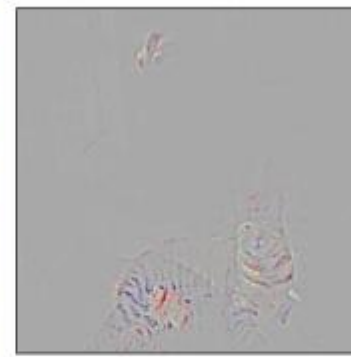
(a) Original Image



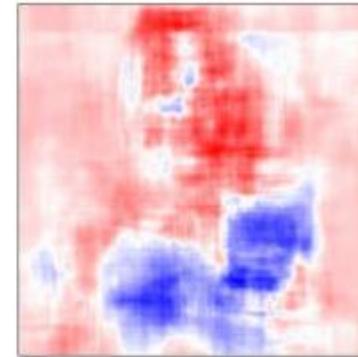
(b) Guided Backprop 'Cat'



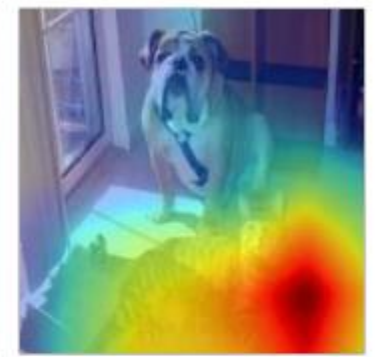
(c) Grad-CAM 'Cat'



(d) Guided Grad-CAM 'Cat'



(e) Occlusion map for 'Cat'



(f) ResNet Grad-CAM 'Cat'



(g) Original Image



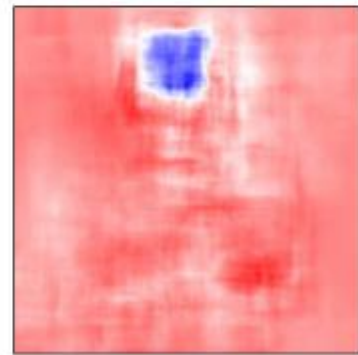
(h) Guided Backprop 'Dog'



(i) Grad-CAM 'Dog'



(j) Guided Grad-CAM 'Dog'

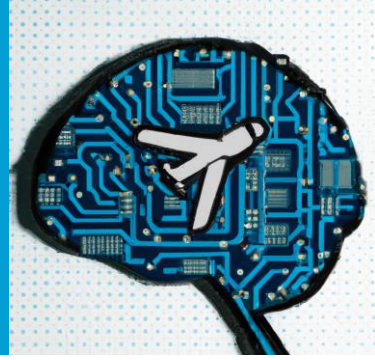


(k) Occlusion map for 'Dog'

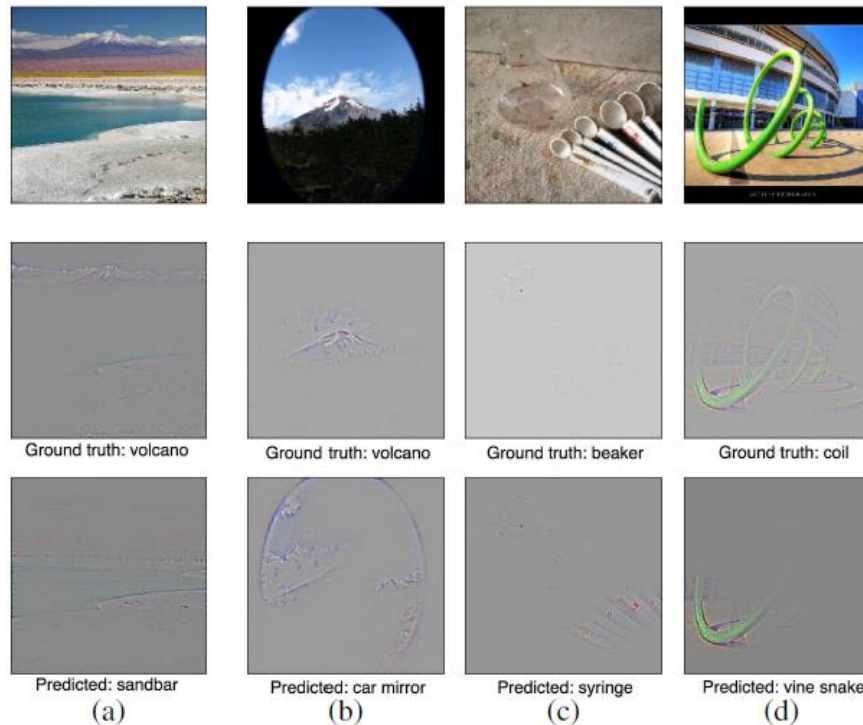


(l) ResNet Grad-CAM 'Dog'

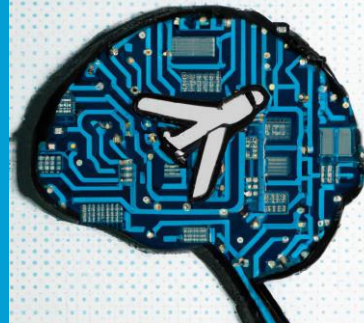
Grad-CAM



Grad-CAM can also be used to analyze failure modes, here of VGG-16.



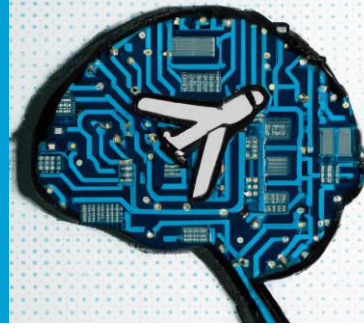
Gradient-based methods



Advantages

Disadvantages

Gradient-based methods



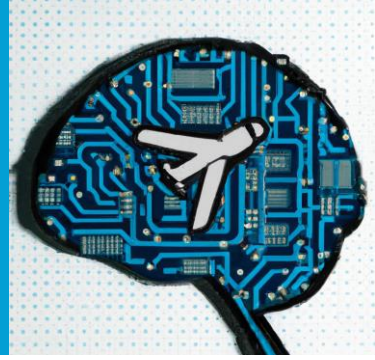
Advantages

- ✓ Visual representations are intuitive.
- ✓ Fast to compute.

Disadvantages

- ✗ Fragility / sensitivity to perturbations
- ✗ What does it really explain? E.g., gradients are often located on edges, making the methods fancy edge detectors.

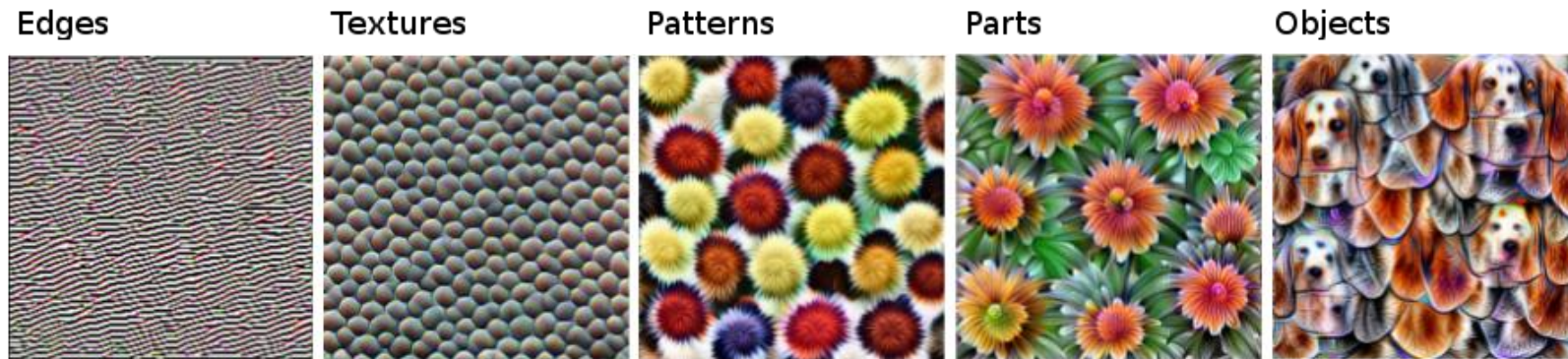
Feature visualization



Feature visualization for a unit:

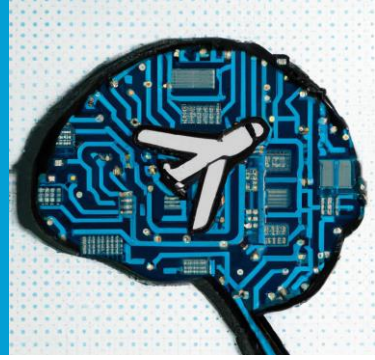
- Finding the input that maximizes the unit's activation

Unit = layer, channel, neuron



Features learned by a convolutional neural network (Inception V1), from lower-level to higher-level convolutional layers.

Feature visualization

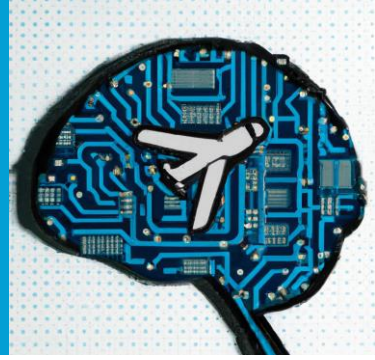


Find an image that maximizes the unit's activation, e.g., for a single channel:

$$I^* = \operatorname{argmax}_I \sum_{x,y} h(l, c, x, y)$$

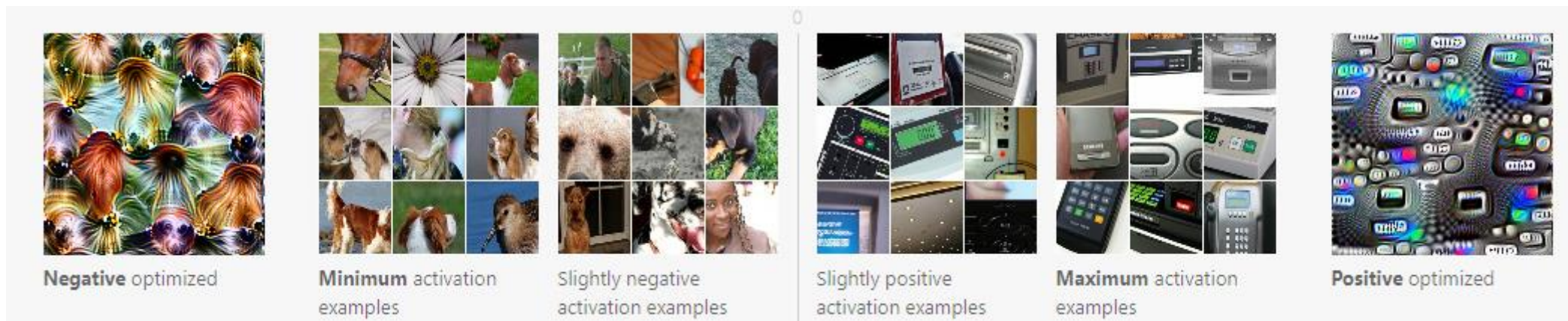
Where l is the layer, c the channel, and x, y the spatial coordinates.

Feature visualization

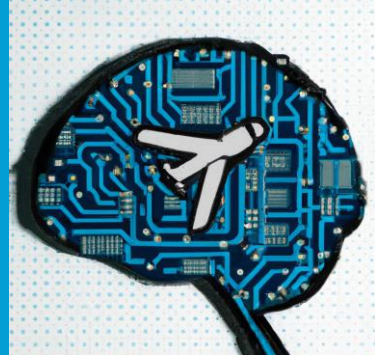


Approaches:

- Search in the data for the highest activation
- Generate images that activate the unit, e.g., starting from noisy images.



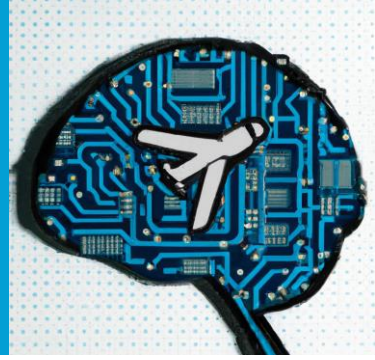
Feature visualization



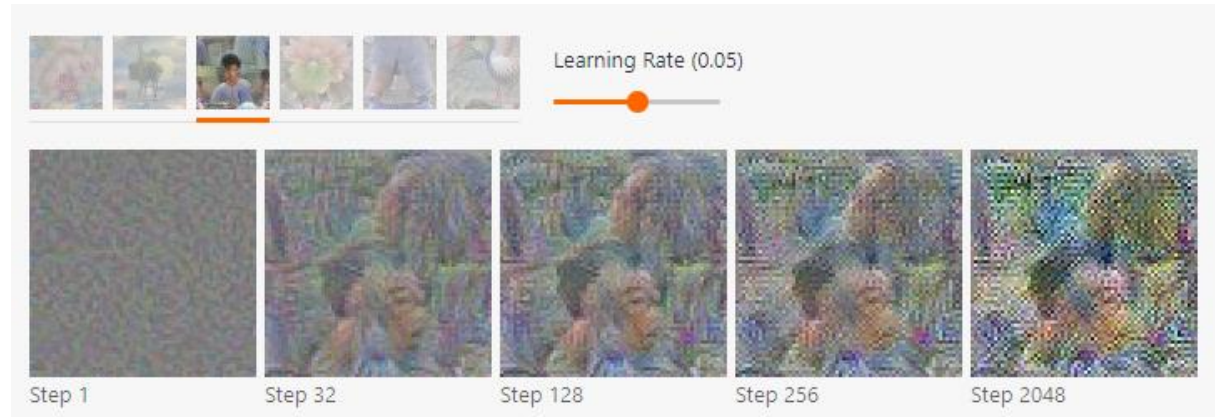
The challenge with optimization lies in high-frequency components:



Feature visualization



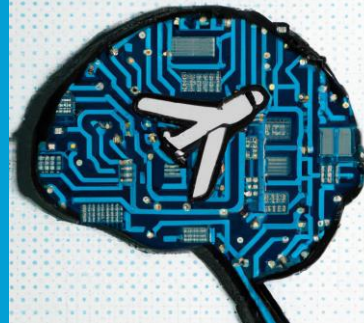
The challenge with optimization lies in high-frequency components:



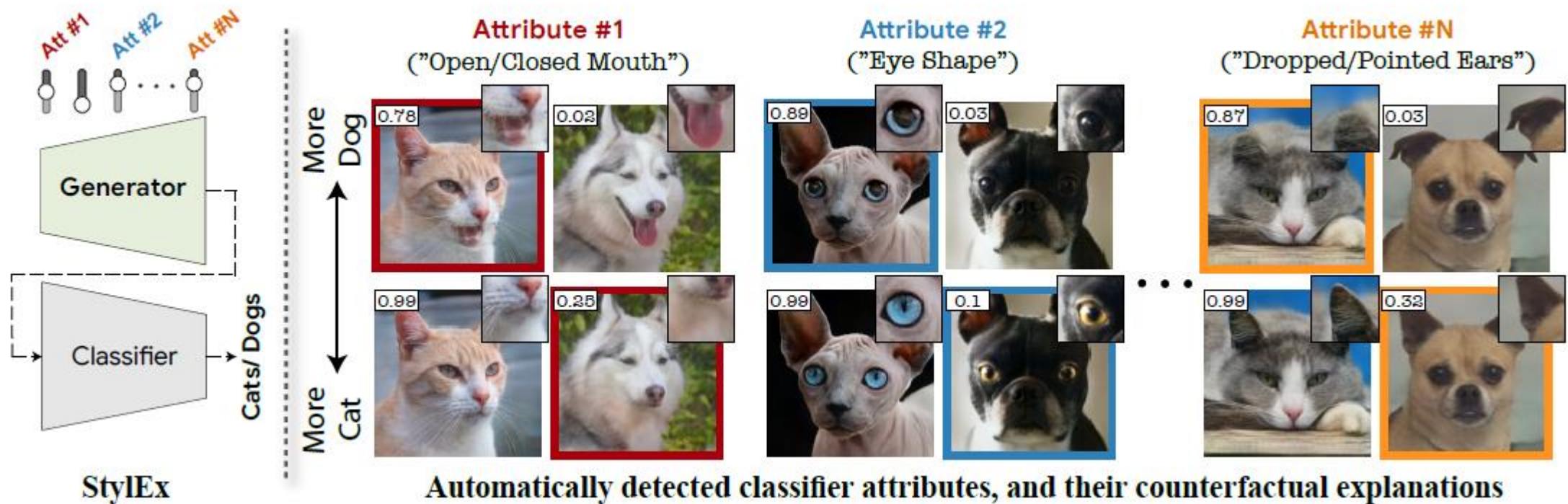
Approaches to tackle this problem:

- Frequency Penalization
- Transformation Robustness
- Learned priors = use generative models

Style-X

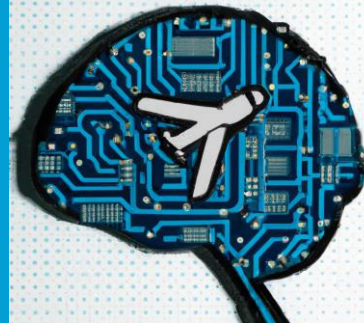


Train a StyleGAN to reveal different attributes contributing to classifications.



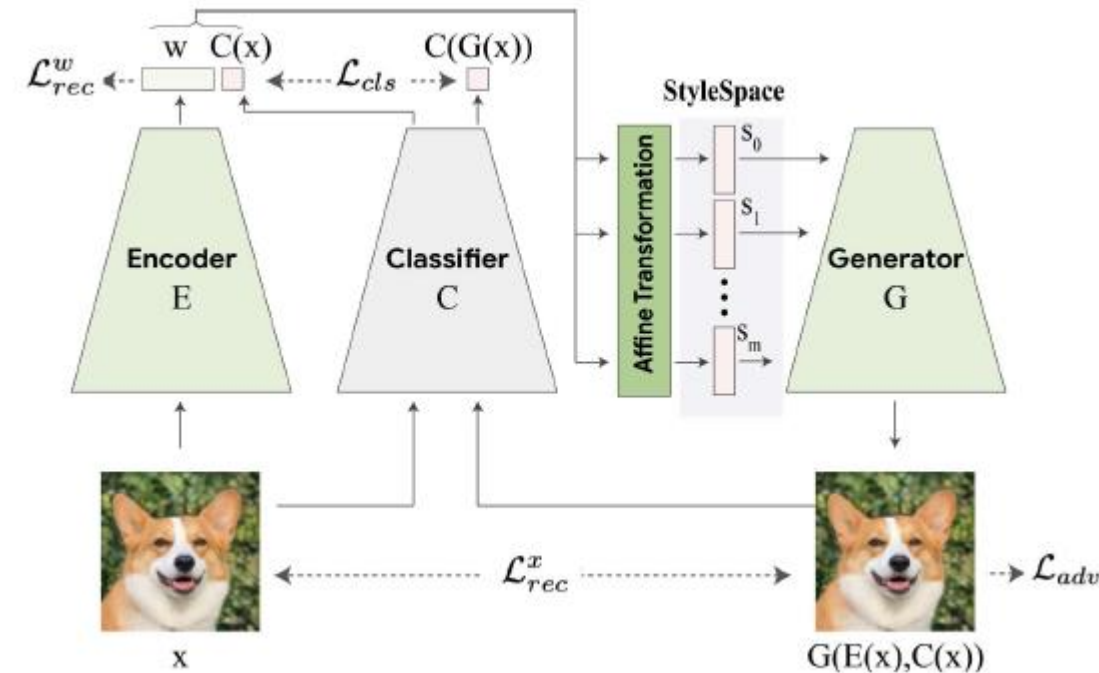
Lang, O., Gandelsman, Y., Yarom, M., Wald, Y., Elidan, G., Hassidim, A., ... & Mosseri, I. (2021). Explaining in style: Training a gan to explain a classifier in stylespace. In Proceedings of the IEEE/CVF International Conference on Computer Vision (pp. 693-702).

Style-X

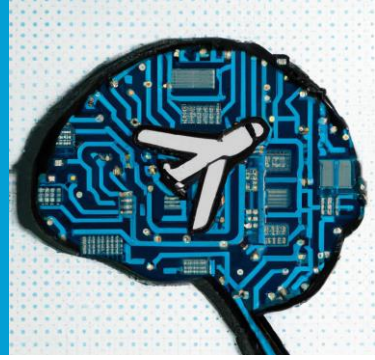


Train a StyleGAN to reveal different attributes contributing to classifications.

1. Train an encoder and generator.
2. Find features in style space (w) that influence the classification most.
3. Generate image-specific examples.



Feature visualization



Advantages

✓ Allow you to visualize the concepts learned in different network elements in an interpretable way.

Disadvantages

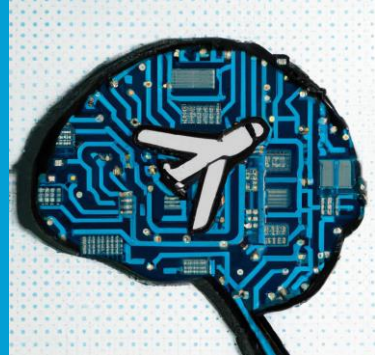


“Dumbbell” feature



<https://research.google/blog/inceptionism-going-deeper-into-neural-networks/>

Feature visualization



Advantages

✓ Allow you to visualize the concepts learned in different network elements in an interpretable way.

Disadvantages

✗ Can be misleading, as optimizing random directions in latent space also give interpretable concepts.

“Dumbbell” feature



<https://research.google/blog/inceptionism-going-deeper-into-neural-networks/>

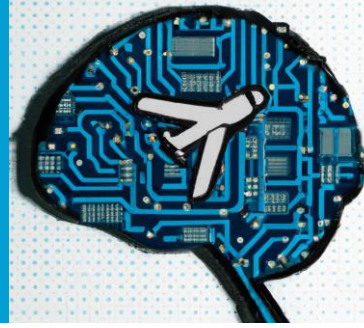
Optimizing random latent space directions



mixed3a, random direction *mixed4c, random direction* *mixed4d, random direction* *mixed5a, random direction*

<https://distill.pub/2017/feature-visualization/>

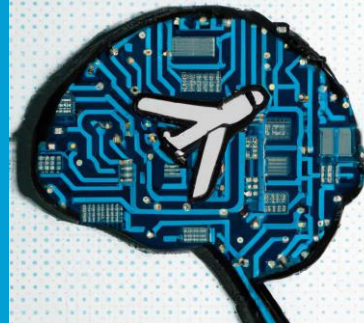
Analyzing latent space information



Dimensionality reduction represents an alternative method for analyzing the latent space information.

Typically, the samples in the lower dimensions are colored according to a variable of interest (class, regression value, etc.)

Analyzing latent space information



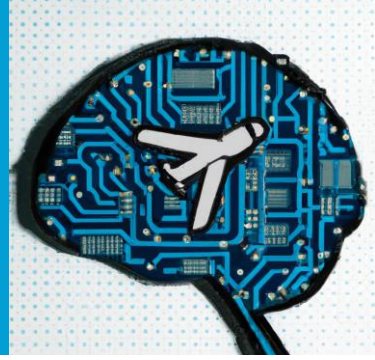
Dimensionality reduction represents an alternative method for analyzing the latent space information.

Typically, the samples in the lower dimensions are colored according to a variable of interest (class, regression value, etc.)

Often-used dimensionality reduction algorithms:

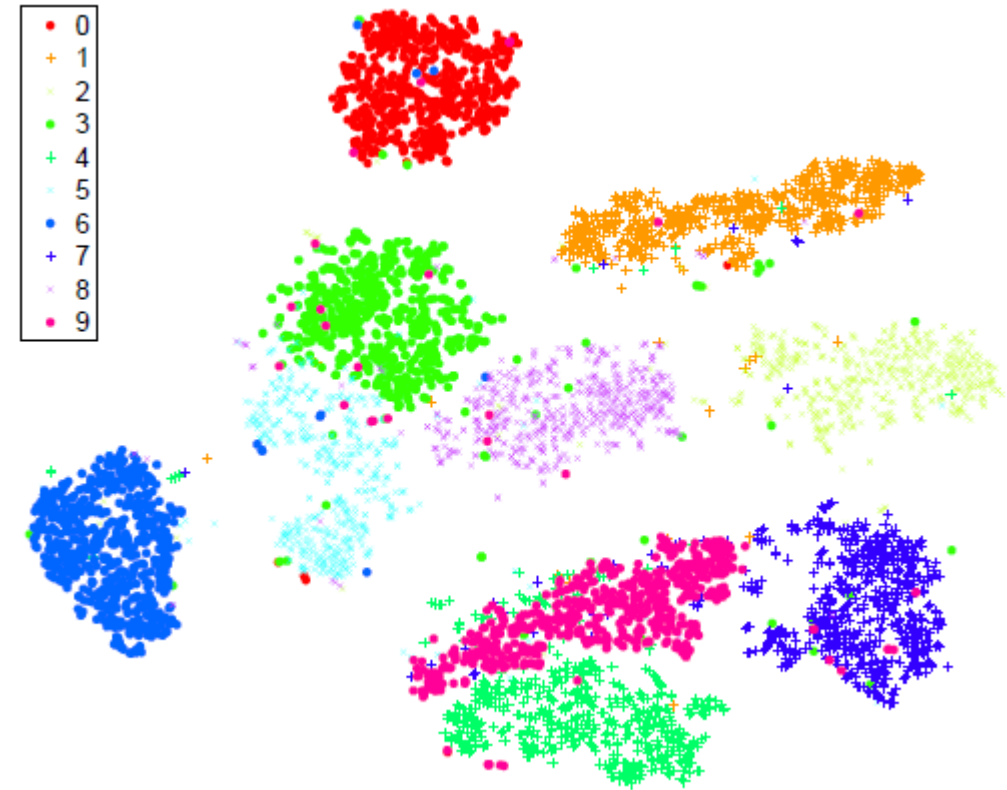
- t-Distributed Stochastic Neighbor Embedding (t-SNE)
- Uniform Manifold Approximation and Projection (UMAP)

t-SNE

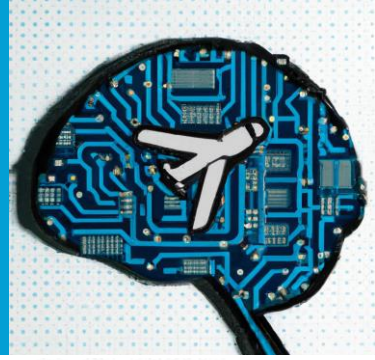


Performs dimensionality reduction by trying to maintain distances in the original higher-dimensional space in the low-dimensional space.

Visualization of the t-SNE embedding in 2 dimensions of the MNIST dataset (samples of dimension 28x28 pixels).

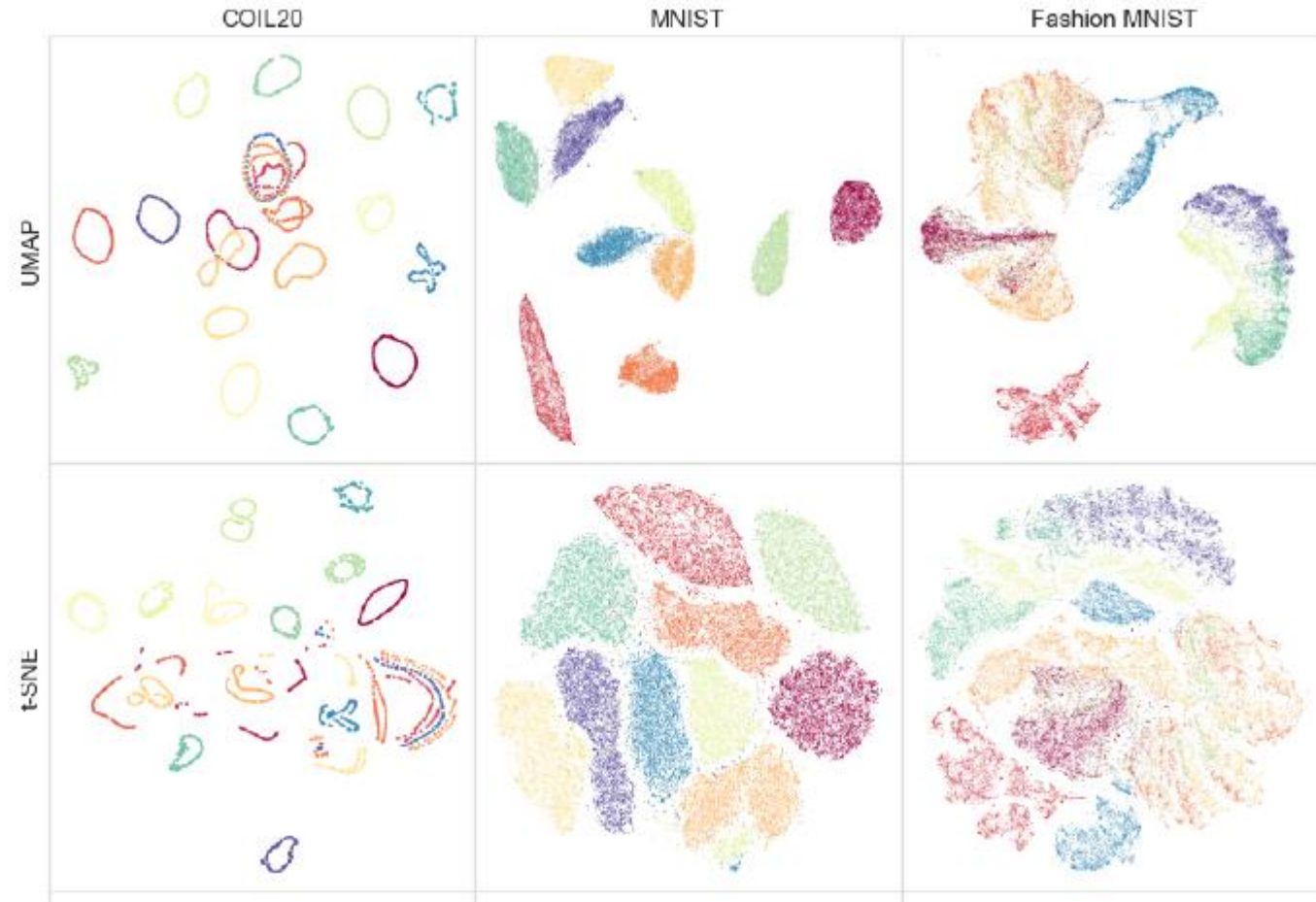


U-MAP



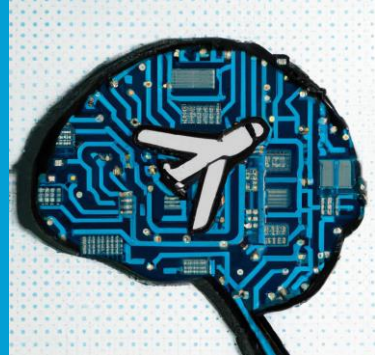
Competitive with t-SNE.

Tries to preserve more of the global structure in the high-dimensional space.



McInnes, L., Healy, J., & Melville, J. (2018). Umap: Uniform manifold approximation and projection for dimension reduction. arXiv preprint arXiv:1802.03426.

“Manual” input manipulation



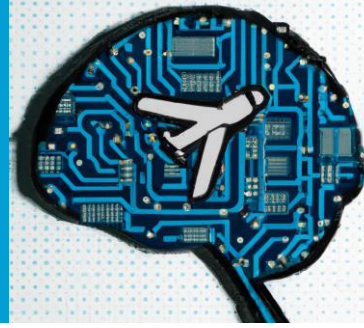
The strategy employed by a deep neural network can be studied by manipulating the inputs.

This “black box” approach is very similar to how cognitive psychology studies are set up.



“Black box” approach

Case study: Monocular depth estimation

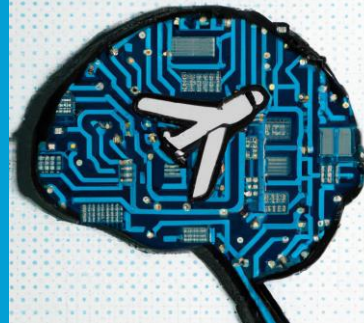


Can you see depth in the picture on the right?

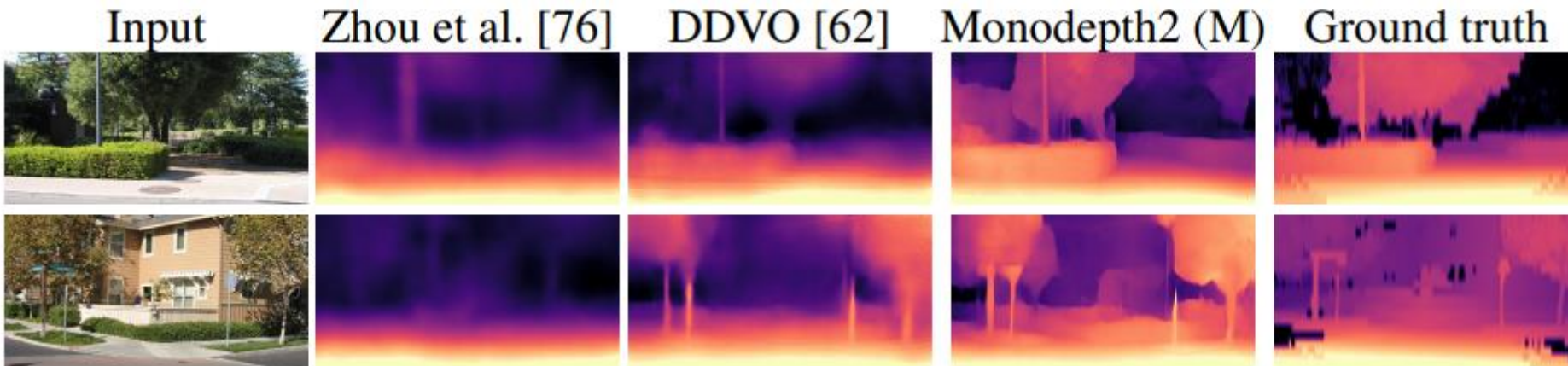
What visual cues do we use?



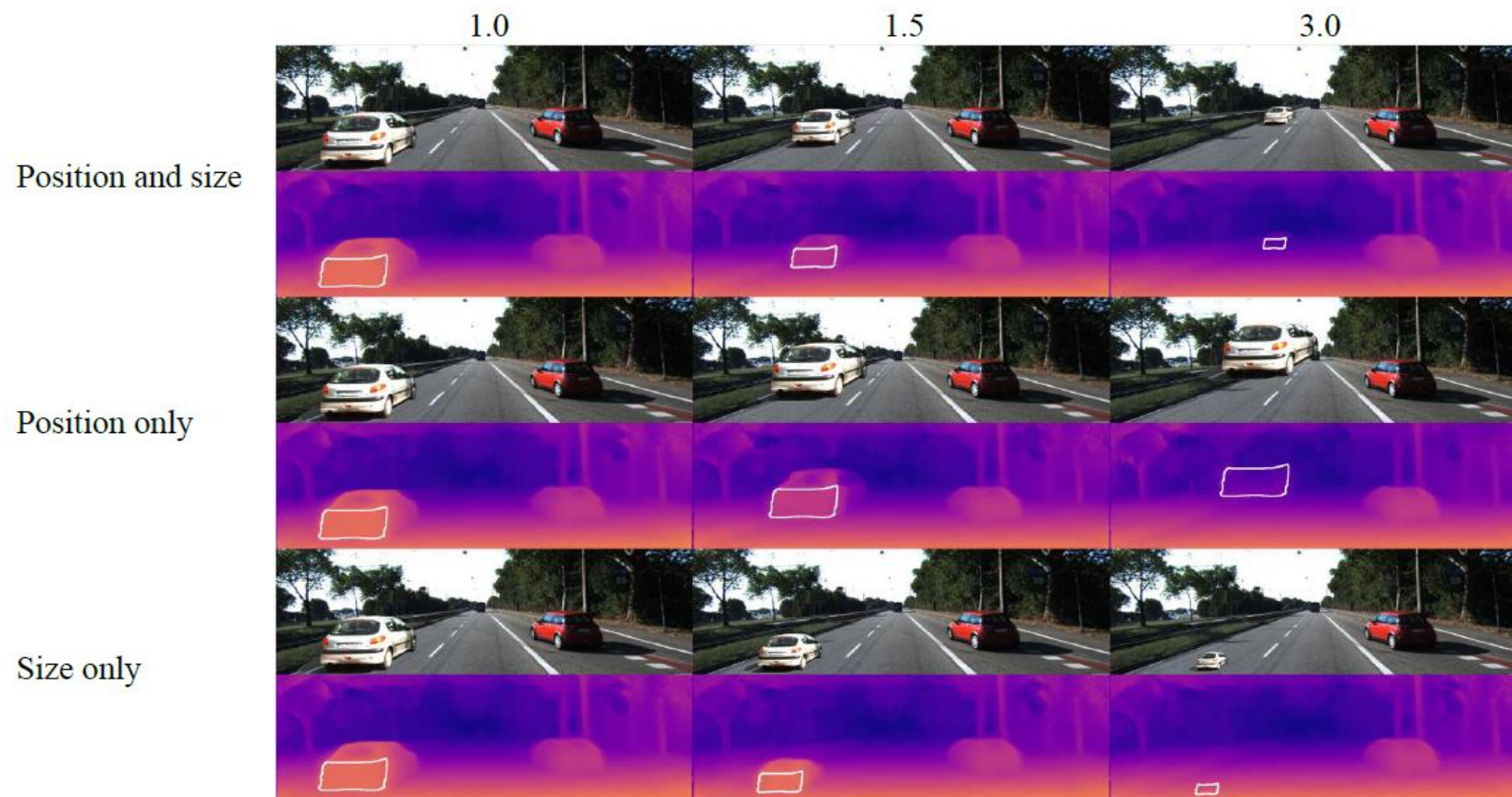
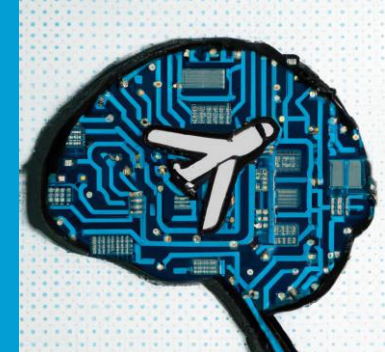
Monocular depth estimation with DNNs



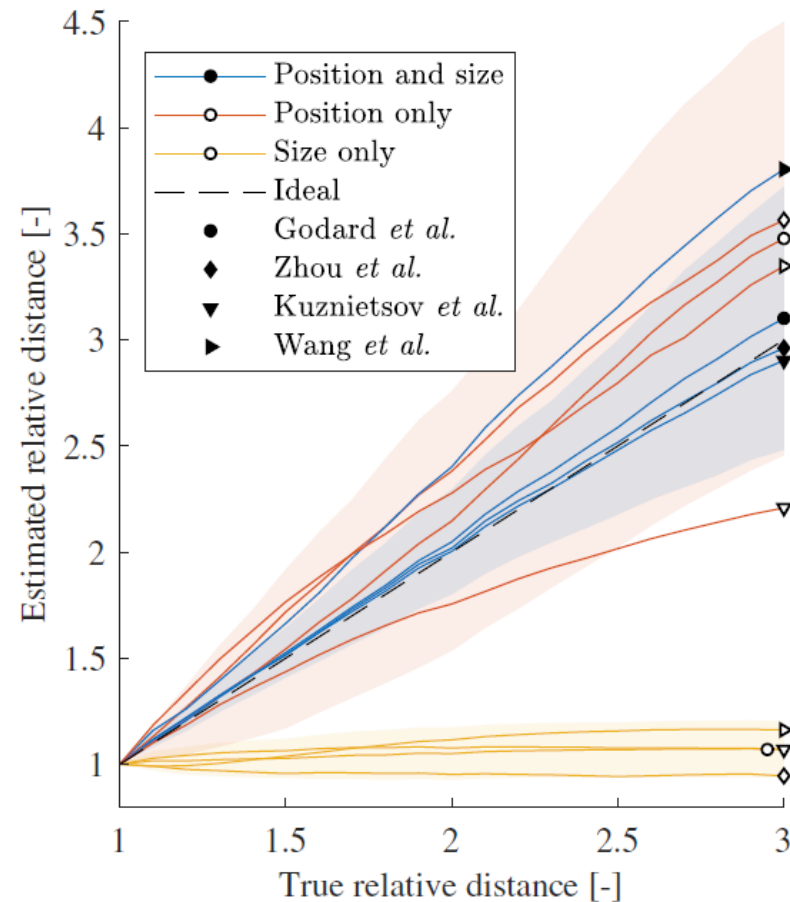
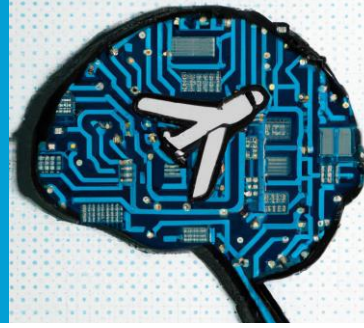
Deep neural networks can also be trained to see monocular depth.



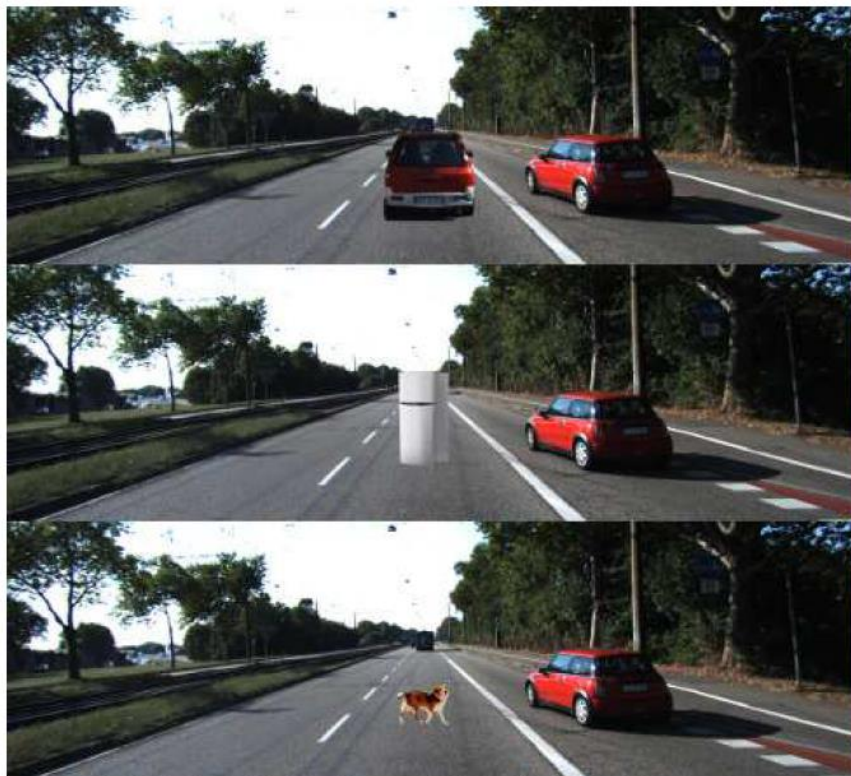
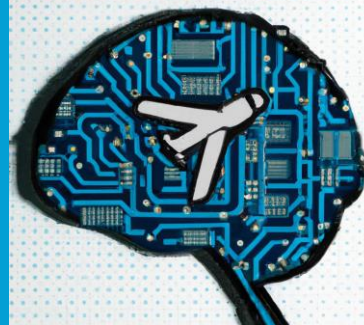
How do deep nets see monocular depth?



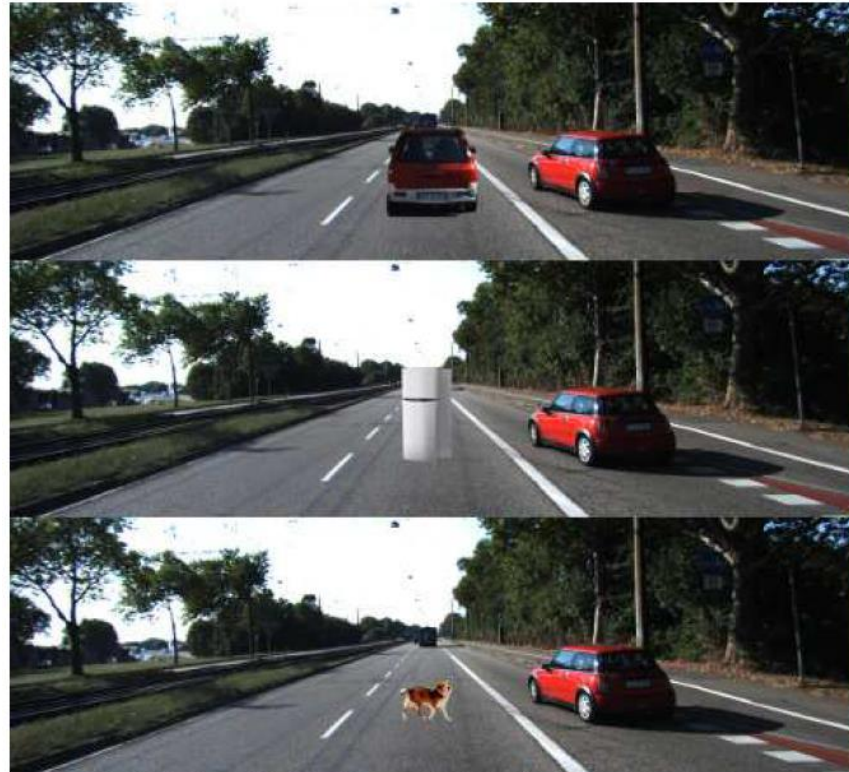
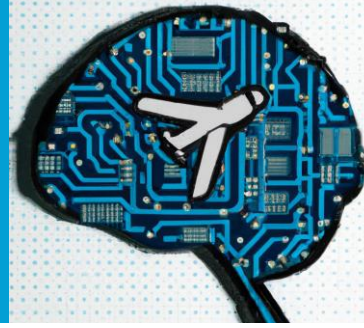
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How do deep nets see monocular depth?

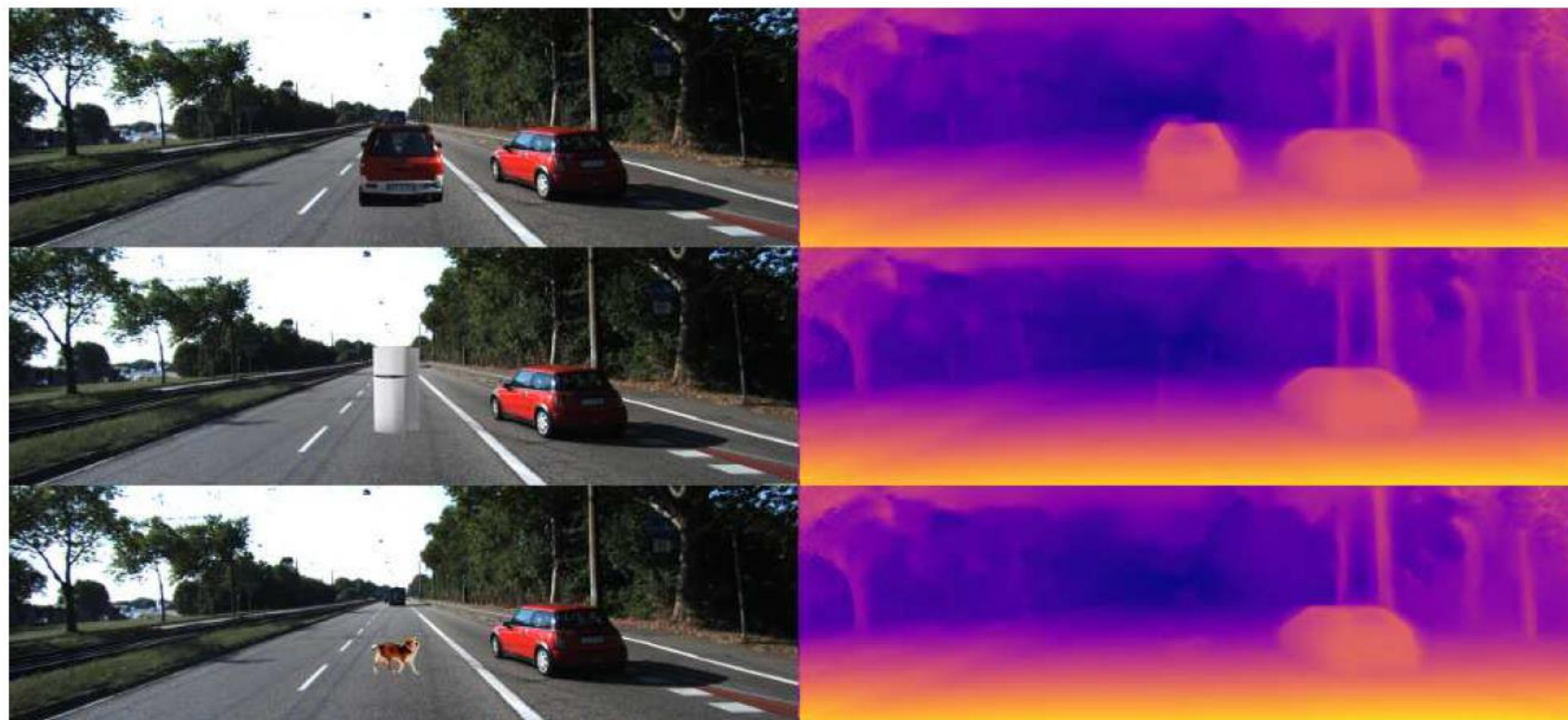
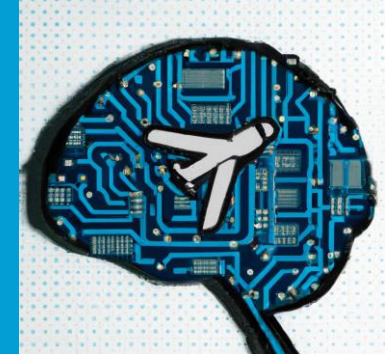


How do deep nets see monocular depth?

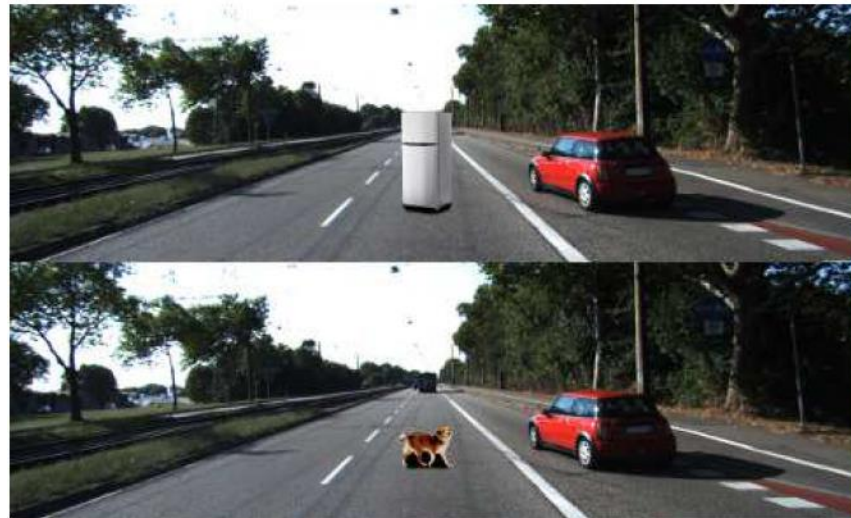
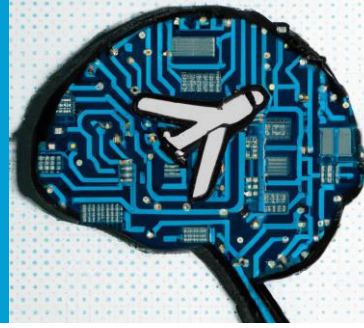


What will happen if we give these images to MonoDepth?

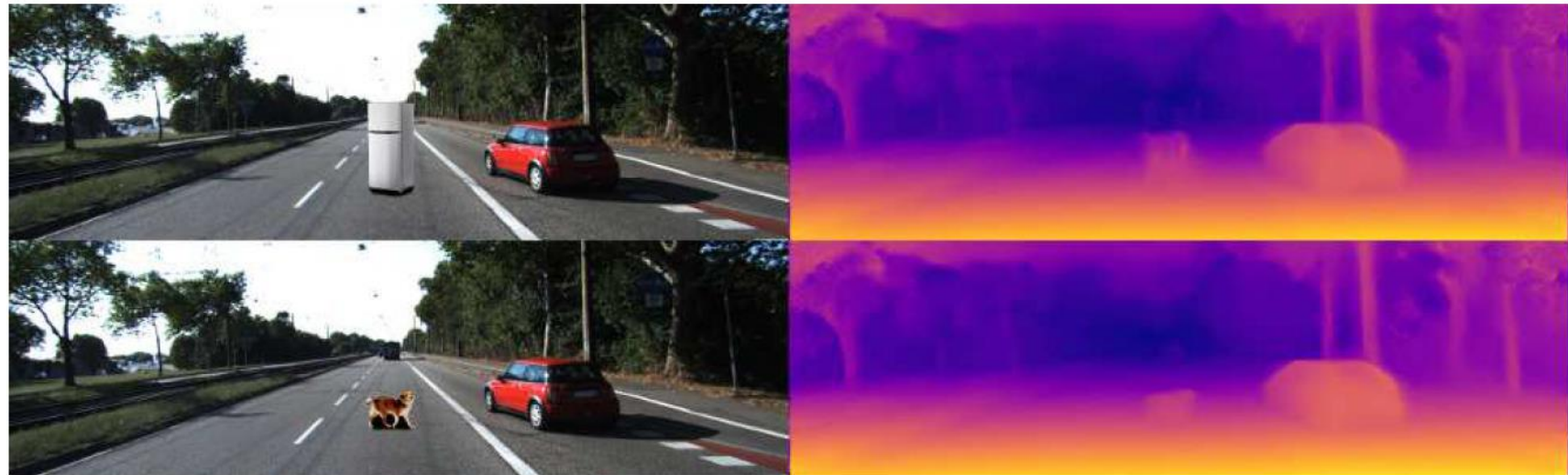
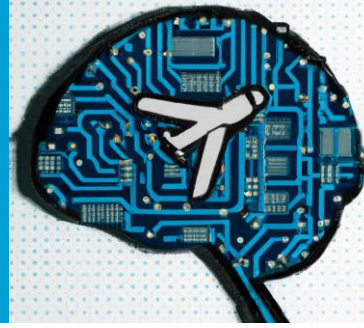
How do deep nets see monocular depth?



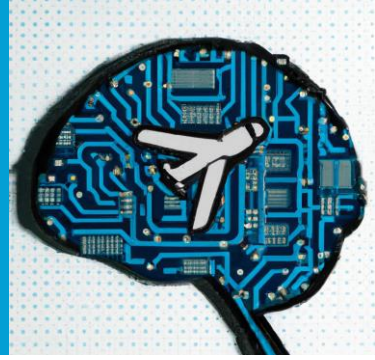
How do deep nets see monocular depth?



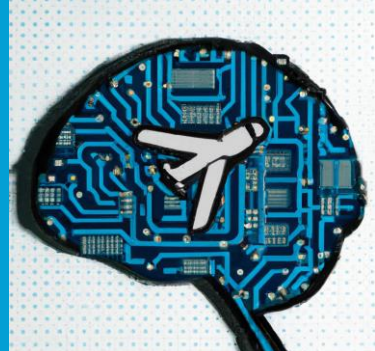
How do deep nets see monocular depth?



Conclusion

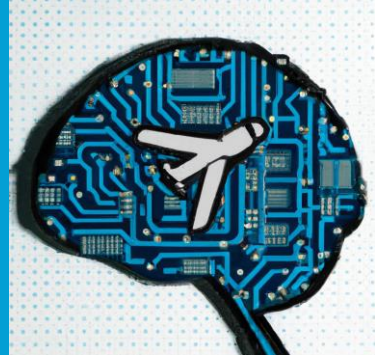


- Understanding deep neural networks is important for building trust in AI systems
- When inputs consist of clearly understandable – and quite decorrelated – input features, methods are well established
- There is a growing literature on understanding deep neural networks with more complex feature representations, but they leave behind some dissatisfaction as to the level of understanding.



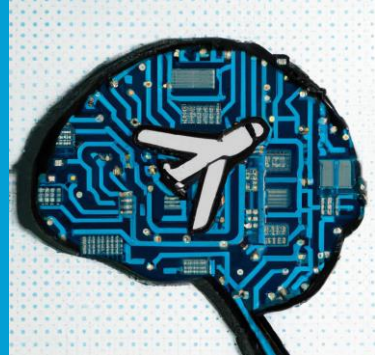
Quiz

Question 1



Why does Grad-CAM typically lead to quite blurry attention areas?

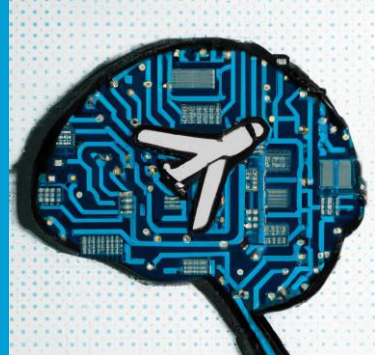
Question 1



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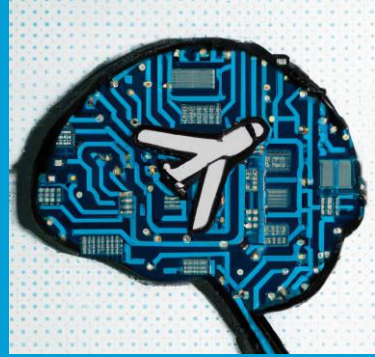
- A. Because Grad-CAM uses gradients from deep layers that are lower in spatial resolution, making the attention maps coarser.
- B. Because Grad-CAM computes the average of gradients over all pixels, resulting in less precise attention localization.
- C. Because Grad-CAM is secretly a fan of impressionist art and prefers everything to be a bit blurry for aesthetic reasons.
- D. Because Grad-CAM focuses on all neurons in the network, causing overlapping activations that blur the attention areas.

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Questions?

Analyzing networks

